MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

DEMULTIPLEXER, ANALOG, KEY CODE VGS

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic demultiplexer module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 2.0 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be VGS.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

* 3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/182-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage of 5.0 ±0.5 V dc and ±15.0 ±1.5 V dc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

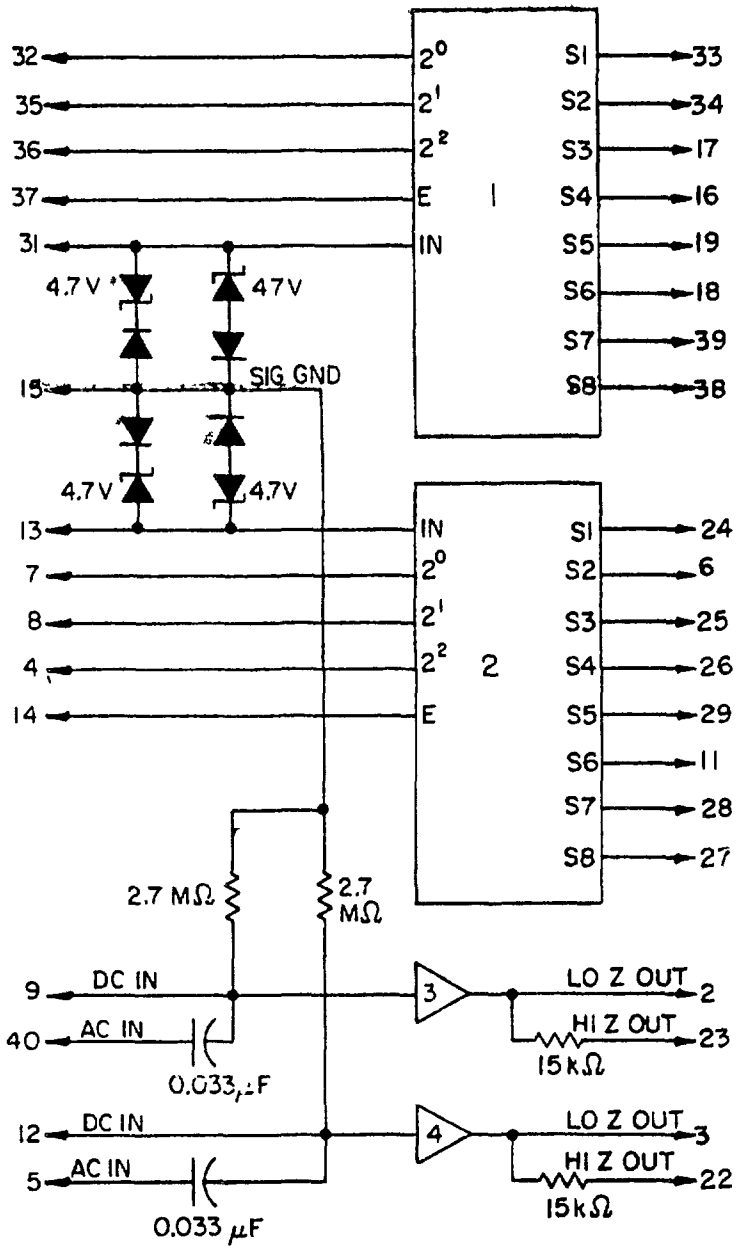
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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Circuits 1 and 2 - Demultiplexers
Circuits 3 and 4 - Voltage follower amplifiers
IN - Analog signal input
2^0, 2^1, 2^2 - Select } Logic inputs
E - Enable }
S1-S8 - Selected analog output channels.

*FIGURE 1. VGS module function diagram, demultiplexer.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A16	V_S stress, ON resistance check	III	Q	0.5	1.0	0.5	1.0	0.5	1.0	0.5	1.0	V rms
A17,A18	V_S stress, offset voltage	III	Q	-15	15	-15	15	-15	15	-15	15	mV
B1,B3	Output short circuit current, voltage follower, negative input	IV	S	-3	-30	-3	-30	-3	-30	-3	-30	mA
B2,B4	Output short circuit current, voltage follower, positive input	IV	S	3	30	3	30	3	30	3	30	mA
B5,B6	Output short circuit current, demultiplexer	IV	S	4.29	15.7	4.29	15.7	4.29	15.7	4.29	15.7	mA
C	Isolation	V	S	10	∞	1	∞	10	∞	1	∞	mΩ
D1	Power supply current, +15 V	VI	S	1	11	1	12	1	12	1	12	mA
D2	Power supply current, -15 V	VI	S	5	23	5	25	5	25	5	25	mA
D3	Power supply current, 5 V	VI	S	5	16	5	18	5	18	5	18	mA
E1,E2	Supply voltage rejection ratio check	VII	S	0	316	0	316	0	316	0	316	µV
F1-F3	Power supply transient amplitude	VIII	Q	0	300	0	300	0	300	0	300	mV
G1-G4	Slew rate check	VIII	X	10	600	10	600	10	600	10	600	ns
H1,H3,H5,H7, H9,H11,H13, H15	Input current, low level	IX	X	0.1	1.2	0.1	1.2	0.1	1.73	0.1	1.73	mA
H2,H4,H6,H8, H10,H12,H14, H16	Input current, high level	IX	X	0	-350	0	-350	0	-350	0	-350	µA
J1-J16	ON resistance check, V_I = 100 mV	X	X	75	100	75	100	71	100	71	100	mV
J17-J32	ON resistance check, V_I = 4 V	X	X	3.94	4.04	3.94	4.04	3.94	4.04	3.94	4.04	V
K1-K16	Large signal voltage response	XI	X	2.75	2.83	2.75	2.83	2.75	2.83	2.75	2.83	V rms
K17-K32	Distortion	XI	X	0	1	0	1	0	1	0	1	%

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
L1-L18	Large signal channel-to-channel attenuation	XII	Q	0	5	0	5	0	5	0	5	mV rms
M1-M8	Crosstalk check	XIII	Q	0.0	1.8	0.0	1.8	0.0	1.8	0.0	1.8	mV rms
M9-M16	Crosstalk check	XIII	Q	0	895	0	895	0	895	0	895	µV rms
N1,N2	Input clamp diode voltage	XIV	X	4.4	5.7	4.4	5.7	4.4	6.0	4.4	6.0	V
N3,N4	Input clamp diode voltage	XIV	X	-4.4	-5.7	-4.4	-5.7	-4.3	-6.0	-4.3	-6.0	V
P1,P2	Capacitance	XV	Q	5	100	5	100	5	100	5	100	pF
P3-P26	Capacitance	XV	Q	5	25	5	25	5	25	5	25	pF
Q1,Q2	Buffer coupling capacitance	XVI	X	0.026	0.040	0.026	0.040	0.026	0.040	0.026	0.040	µF
Q3,Q4	Buffer coupling dissipation factor	XVI	X	0	5	0	5	0	5	0	5	%
R1-R32	Leakage current	XVII	X	-5	100	-5	100	-5	500	-5	500	nA
R33-R36	Leakage current	XVII	X	0	10	0	15	0	25	0	30	µA
S1-S16	Switching time, t_{ON}	XVIII	S	0.1	3.0	0.1	3.5	0.1	3.5	0.1	3.5	µs
S17-S32	Switching time, t_{OFF}	XVIII	S	0.1	2.0	0.1	2.5	0.1	2.5	0.1	2.5	µs
S33,S34	Switching time, t_{OPEN}	XVIII	S	0.05	3.00	0.05	3.00	0.05	3.00	0.05	3.00	µs
T1,T2	Large signal voltage response	XIX	X	6.9	7.1	6.9	7.1	6.9	7.1	6.9	7.1	V rms
T3,T4	Distortion	XIX	X	0	1	0	1	0	1	0	1	%
U1,U2	Low frequency response	XX	X	6.93	7.07	6.93	7.07	6.93	7.07	6.93	7.07	V rms
U3,U4	DC gain	XX	X	6.94	7.06	6.94	7.06	6.94	7.06	6.94	7.06	V
V1,V2	DC offset voltage	XXI	X	-10	10	-10	10	-10	10	-10	10	mV
W1,W2	Output resistance	XXII	X	14.85	15.15	14.7	15.3	14.7	15.3	14.7	15.3	kΩ

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
W3,W4	Input resistance	XXII	X	2.43	2.97	2.43	2.97	2.43	2.97	2.00	3.00	MΩ
X1	Power supply filter capacitance	XXIII	S	1.65	2.75	1.65	2.75	1.65	2.75	1.65	2.75	µF
X2,X3	Power supply filter capacitance	XXIII	S	0.075	0.125	0.075	0.125	0.075	0.125	0.075	0.125	µF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested du ing qualification testing only.

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4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D, E, G through K, N, and Q through X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition X, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D, E, G through K, N, and Q through X shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D, E, G through K, N, and Q through X shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/182-1 supersedes part number M28787/182. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation	715 mW (total, no signal)
+5.5 V	100 mW
+16.5 V	200 mW
-16.5 V	415 mW
Supply voltage	5 V dc ±10 percent and ±15 V dc ±10 percent

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Analog demultiplexer	
Switch action	Break-before-make
Input voltage (select and enable)	
Low level	-0.3 V minimum, 0.6 V maximum
High level	3.5 V minimum, 5.5 V maximum
Signal input voltage range	
"ON" state	$-5\ V \leq V_I \leq +5\ V$ maximum
"OFF" state	$-10\ V \leq V_I \leq \pm 10\ V$ maximum
ON resistance	
25°C	
$I_I = -100\ \mu A$, $V_I = 0.1\ V$	250Ω maximum
$I_I = -100\ \mu A$, $V_I = 4.0\ V$	600Ω maximum
0°C to 60°C	
$I_I = -100\ \mu A$, $V_I = 0.1\ V$	264Ω maximum
$I_I = -100\ \mu A$, $V_I = 4.0\ V$	600Ω maximum
Load resistance (for $-4\ V \leq V_I \leq 4\ V$ undistorted output)	20 kΩ minimum
Input limiting	Input is clamped at ±5 V with zener diodes
Voltage follower	
Gain	
DC coupled ($DC \leq f \leq 20$ Hz)	1.0 ±0.9 percent
AC coupled ($50\ Hz \leq f \leq 20$ kHz)	1.0 ±1.5 percent
Input impedance	
DC coupled	2.7 MΩ ±10 percent
AC coupled	Series 0.033 µF ±20 percent with 2.7 MΩ ±10 percent to ground
Input voltage	±15 V maximum
Output voltage (R_L = 10 kΩ)	±10 V maximum
Source resistance	10 kΩ maximum
Power supply rejection ratio	70 dB minimum
Slew rate (R_L = 10 kΩ)	16.7 V/µs minimum

* 6.5 Module function. This module contains analog demultiplexers (also usable as multiplexers) and two voltage follower amplifiers.

(a) Each demultiplexer has one input and eight selectable outputs. The outputs are selected via the logic inputs. There are four logic lines; one is an enable line and the other three form a three bit binary word. The binary word will select one of eight outputs. This circuit can also be used as a multiplexer, with eight inputs and one output. One of eight inputs would be selected with the three bit binary word. Logic input levels "L" and "H" correspond to a positive logic low level of ≤0.6 V and high level of ≥3.5 V. If the logic inputs are driven by a single TTL gate, no additional pull-up resistor is required to obtain the 3.5 V minimum high level. The analog signal voltage range is ±5 V in the "on" state. The demultiplexer will block an analog signal with a voltage range of ±10 V in the "off" state. There is no voltage offset through the switch in the "on" state. The module shall provide 64 dB of isolation between the two demultiplexers. On a single demultiplexer there shall be not less than 55 dB of isolation to "off" outputs with a load resistance of 15 kΩ and a frequency of 15 kHz or less.

(b) Each voltage follower has two inputs and two outputs. One input is dc coupled and one input is ac coupled with a low frequency cutoff (-3 dB) of 1.8 Hz. One of the outputs is direct and the other has a 15 kΩ series resistor. The direct coupled output cannot be short circuited without damage if the input is driven from a low impedance source. The output can be short circuited continuously without damage if the input is driven from a source impedance of greater than 2 kΩ.

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6.5.1 Input to output relationship. Each analog demultiplexer has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship. 1/ - 3/

Logic inputs				ON channel
2^0	2^1	2^2	Enable	
L	L	L	H	S1
H	L	L	H	S2
L	H	L	H	S3
H	H	L	H	S4
L	L	H	H	S5
H	L	H	H	S6
L	H	H	H	S7
H	H	H	H	S8
X	X	X	L	All channels off

1/ L = low level ≤0.6 V.
2/ H = high level ≥3.5 V.
3/ X = don't care.

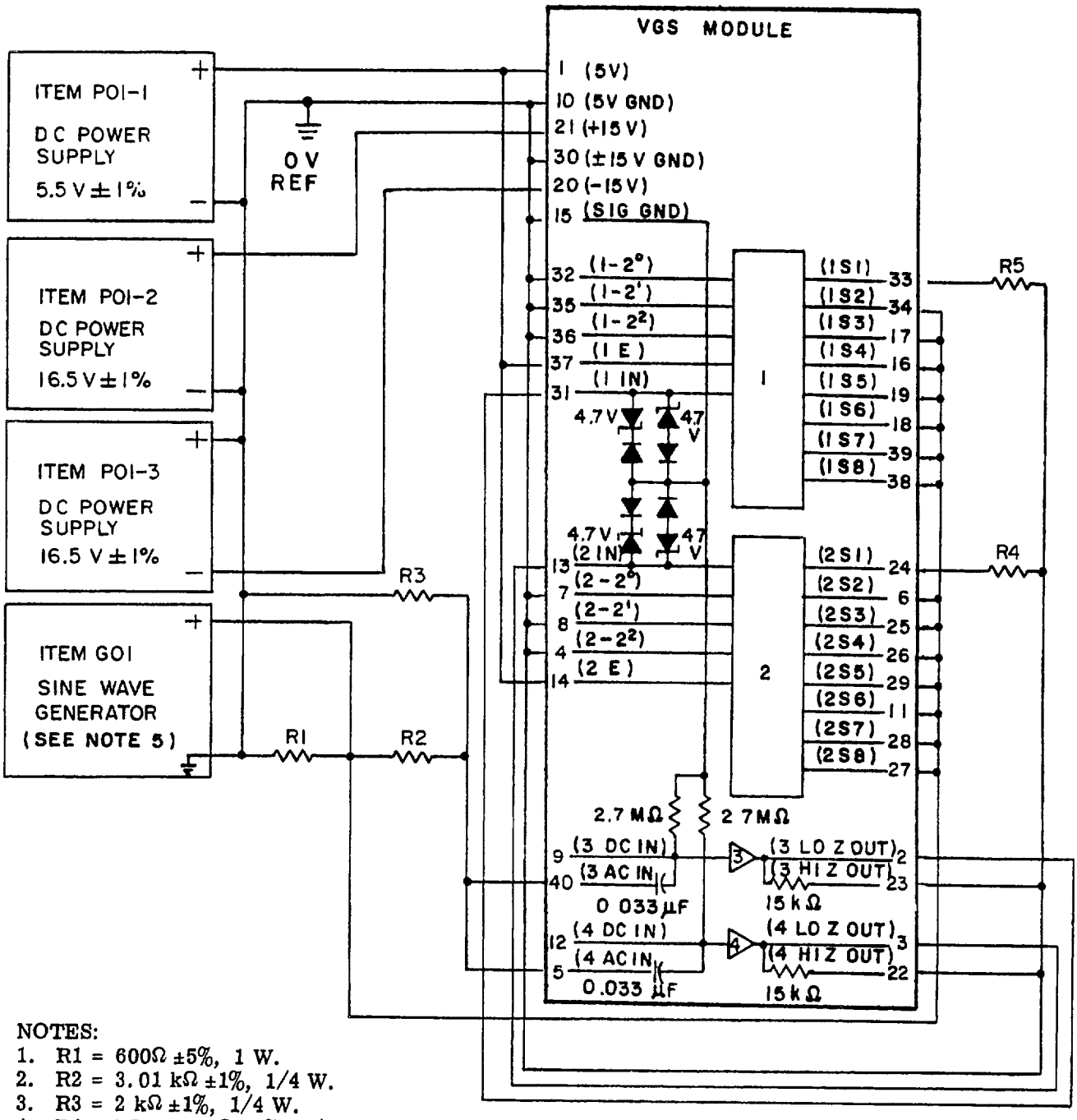
6.5.2 Parallel outputs. Any of the demultiplexer outputs can be connected together. The voltage follower direct coupled outputs must not be paralleled (see 6.5 (b)). The voltage follower outputs which have the 15 kΩ series resistor can be paralleled if desired.

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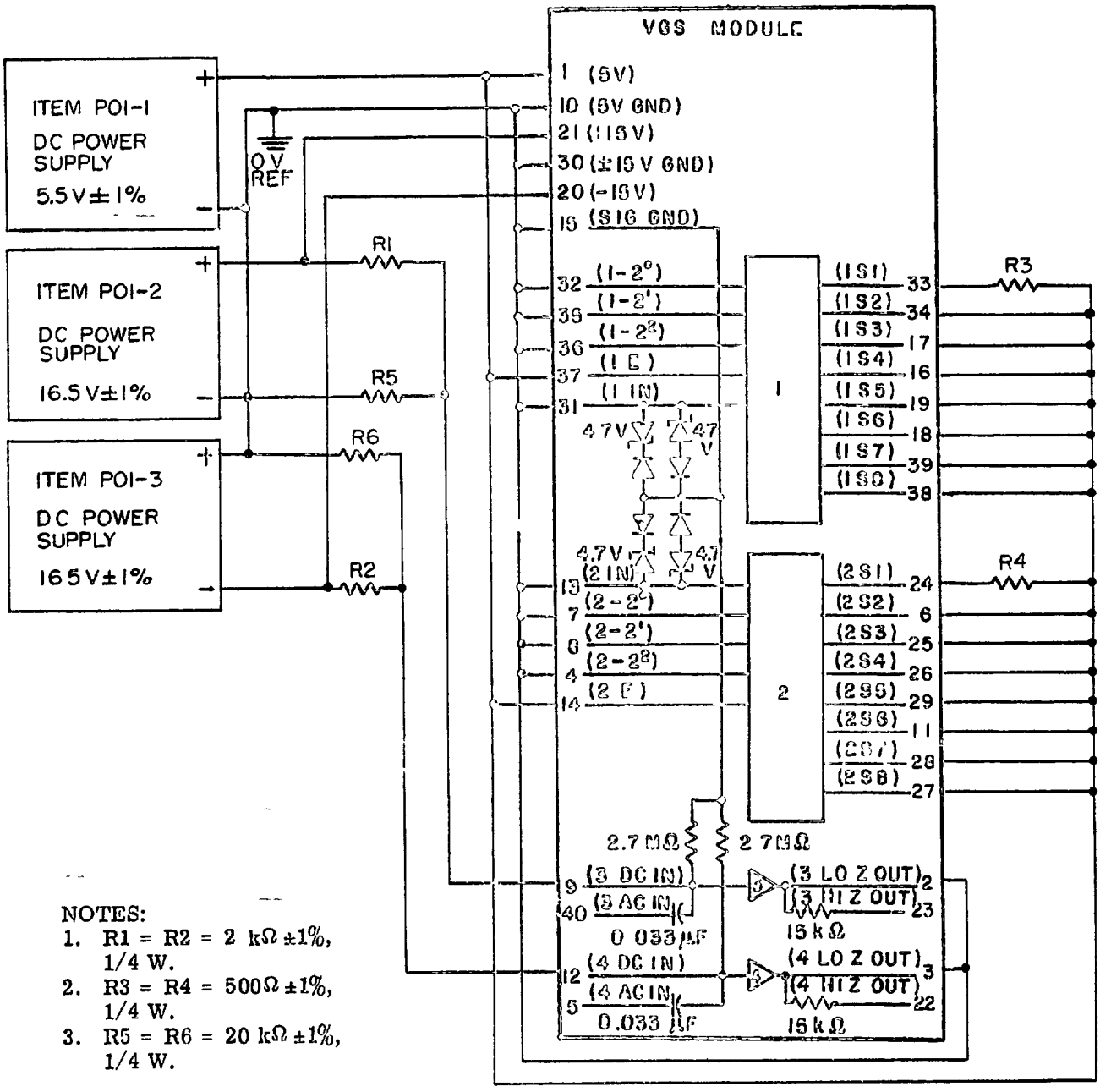
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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = 3.01 kΩ ±1%, 1/4 W.
3. R3 = 2 kΩ ±1%, 1/4 W.
4. R4 = R5 = 20 kΩ ±1%, 1/4 W.
5. Item G01 setup:
   Amplitude = 7V rms ±1%.
   Frequency = 1 kHz ±1%.

*FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = R2 = 2 kΩ ±1%, 1/4 W.
2. R3 = R4 = 500Ω ±1%, 1/4 W.
3. R5 = R6 = 20 kΩ ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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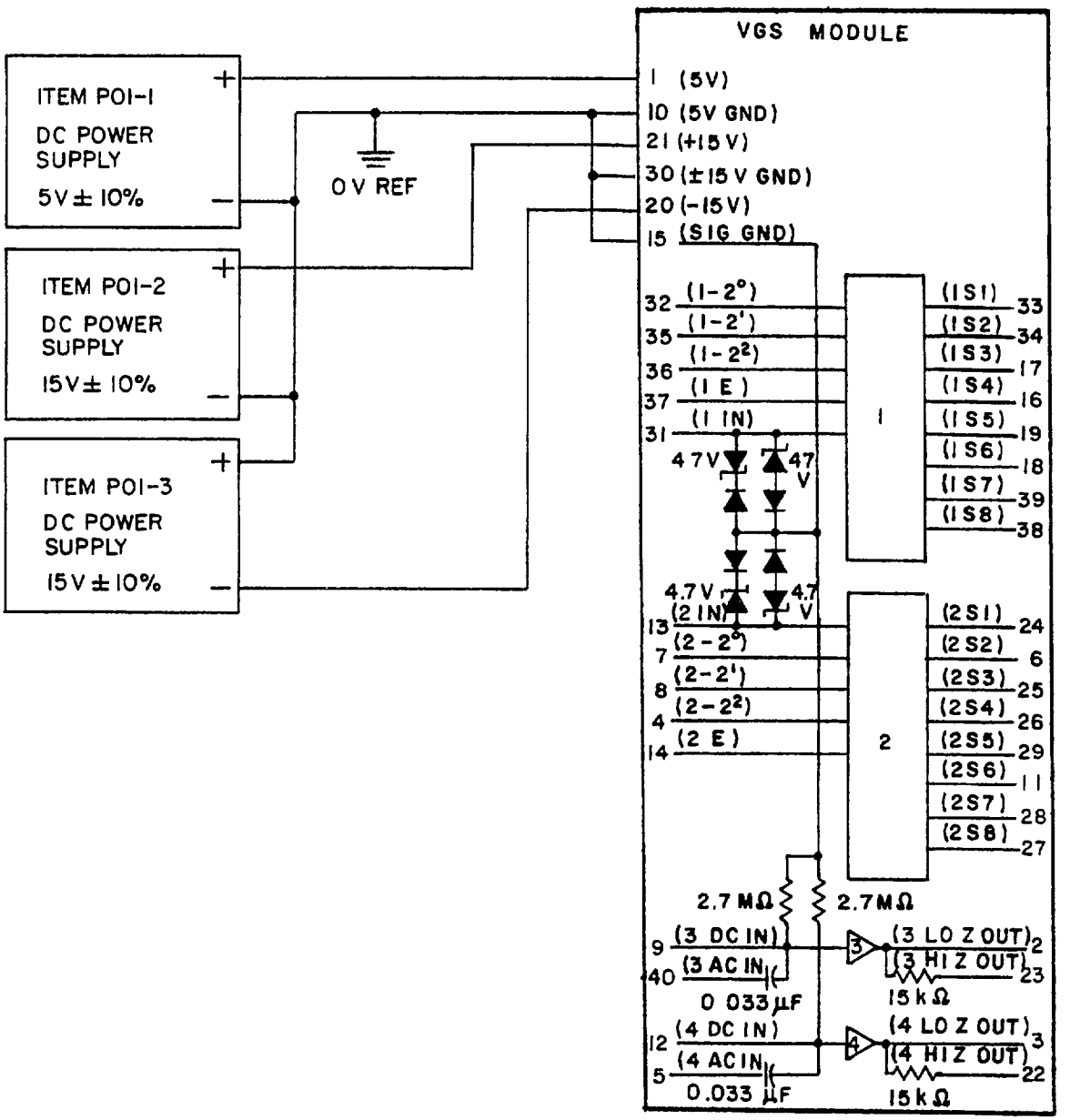


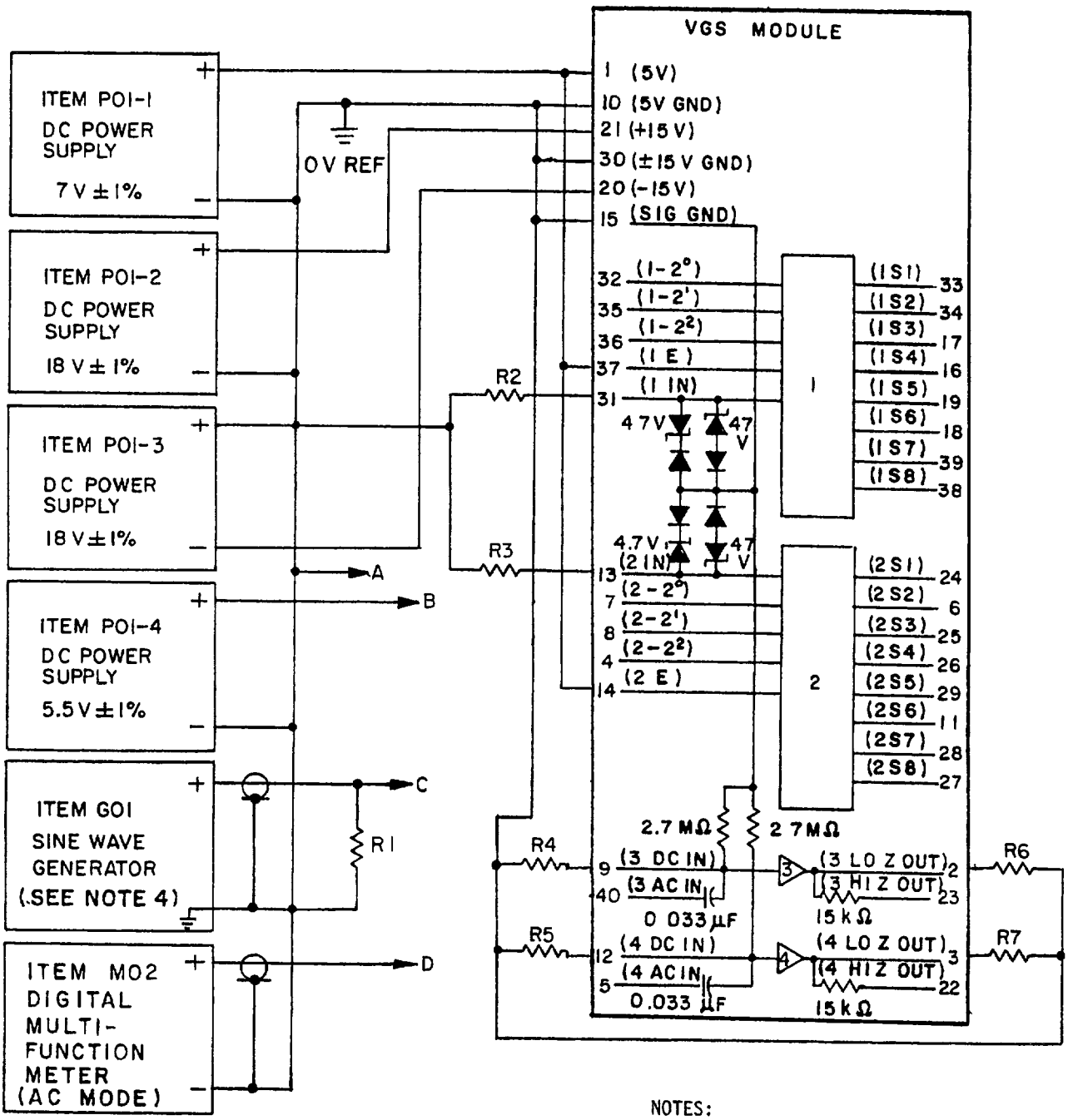
FIGURE 4. Vibration test circuit.

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*TABLE III. V_S stress, ON resistance check and offset voltage test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
A1	1S1 V_S stress, ON resistance check	32,35,36	Open	33	31	0.5	1.0	V rms	5
A2	1S2 "	35,36	32	34	"	"	"	"	"
A3	1S3 "	32,36	35	17	"	"	"	"	"
A4	1S4 "	36	32,35	16	"	"	"	"	"
A5	1S5 "	32,35	36	19	"	"	"	"	"
A6	1S6 "	35	32,36	18	"	"	"	"	"
A7	1S7 "	32	35,36	39	"	"	"	"	"
A8	1S8 "	Open	32,35,36	38	"	"	"	"	"
A9	2S1 "	4,7,8	Open	24	13	"	"	"	"
A10	2S2 "	4,8	7	6	"	"	"	"	"
A11	2S3 "	4,7	8	25	"	"	"	"	"
A12	2S4 "	4	7,8	26	"	"	"	"	"
A13	2S5 "	7,8	4	29	"	"	"	"	"
A14	2S6 "	8	4,7	11	"	"	"	"	"
A15	2S7 "	7	4,8	28	"	"	"	"	"
A16	2S8 "	Open	4,7,8	27	"	"	"	"	"
A17	3LO Z OUT V_S stress, offset voltage	"	Open	Open	2	-15	15	mV	"
A18	4LO Z OUT V_S stress, offset voltage	"	Open	Open	3	-15	15	mV	"

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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 1 kΩ ±1%, 1/4 W.
3. R4-R7 = 10 kΩ ±1%, 1/4 W.
4. Item GOI setup:
   Amplitude = 1 V rms ±0.1%.
   Frequency = 1 kHz ±10%.

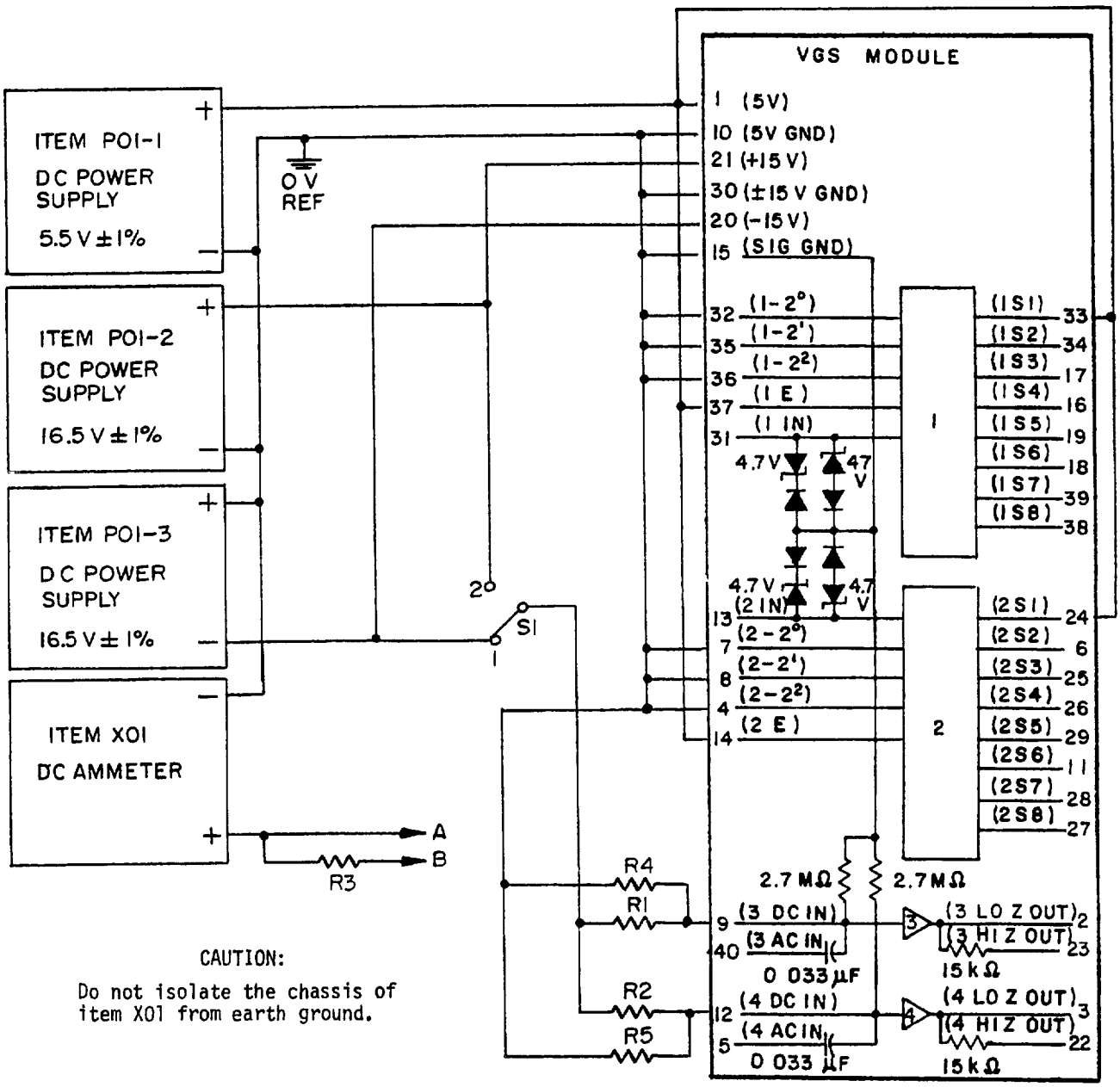
* FIGURE 5. V_S stress, ON resistance check and offset voltage test circuit for table III.

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TABLE IV. Output short circuit current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	3LO Z OUT	Output short circuit current, voltage follower, negative input	2	Open	-3	-30	mA	6
B2	3LO Z OUT	Output short circuit current, voltage follower, positive input	2	"	3	30	"	"
B3	4LO Z OUT	Output short circuit current, voltage follower, negative input	3	"	-3	-30	"	"
B4	4LO Z OUT	Output short circuit current, voltage follower, positive input	3	"	3	30	"	"
B5	1 IN	Output short circuit current, demulti-plexer	Open	31	4.29	15.7	"	"
B6	2 IN	Output short circuit current, demulti-plexer	Open	13	4.29	15.7	"	"

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NOTES:
1. R1 = R2 = 1 kΩ ±1%, 1/4 W.
2. R3 = 348Ω ±1%, 1/4 W.
3. R4 = R5 = 10 kΩ ±1%, 1/4 W.
4. S1 = SPDT switch.
5. Position of S1 shall be as follows:
   Test conditions B1, B3 - Position 1
   Test conditions B2, B4 - Position 2
   Test conditions B5, B6 - Open.

FIGURE 6. Output short circuit current test circuit for table IV.

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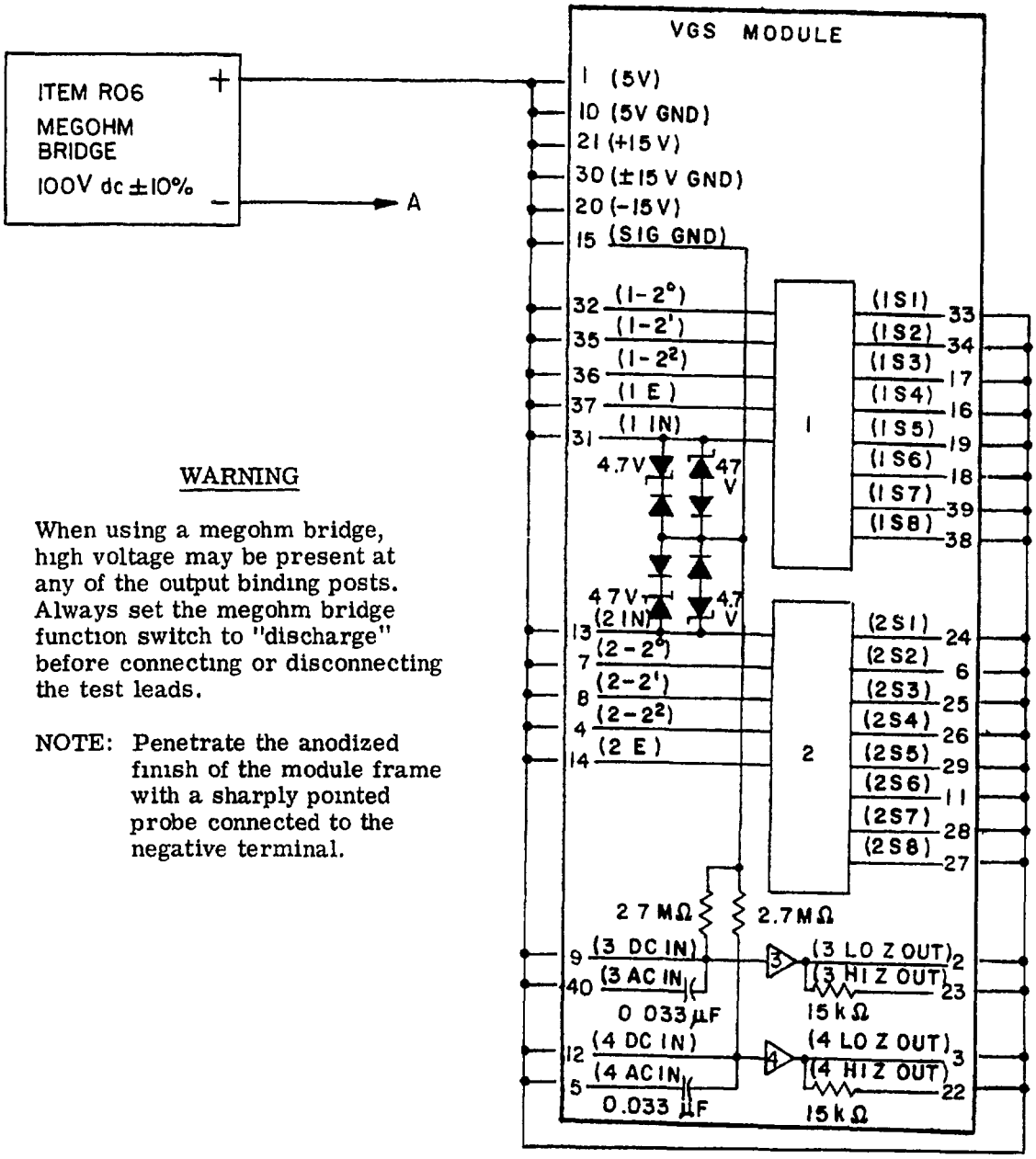
TABLE V. Isolation test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
C	Isolation, frame to module pins	Frame	10	∞	MΩ	7

TABLE VI. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
D1	Power supply current, +15 V	1	11	mA	8
D2	Power supply current, -15 V	5	23	"	"
D3	Power supply current, 5 V	5	16	"	"

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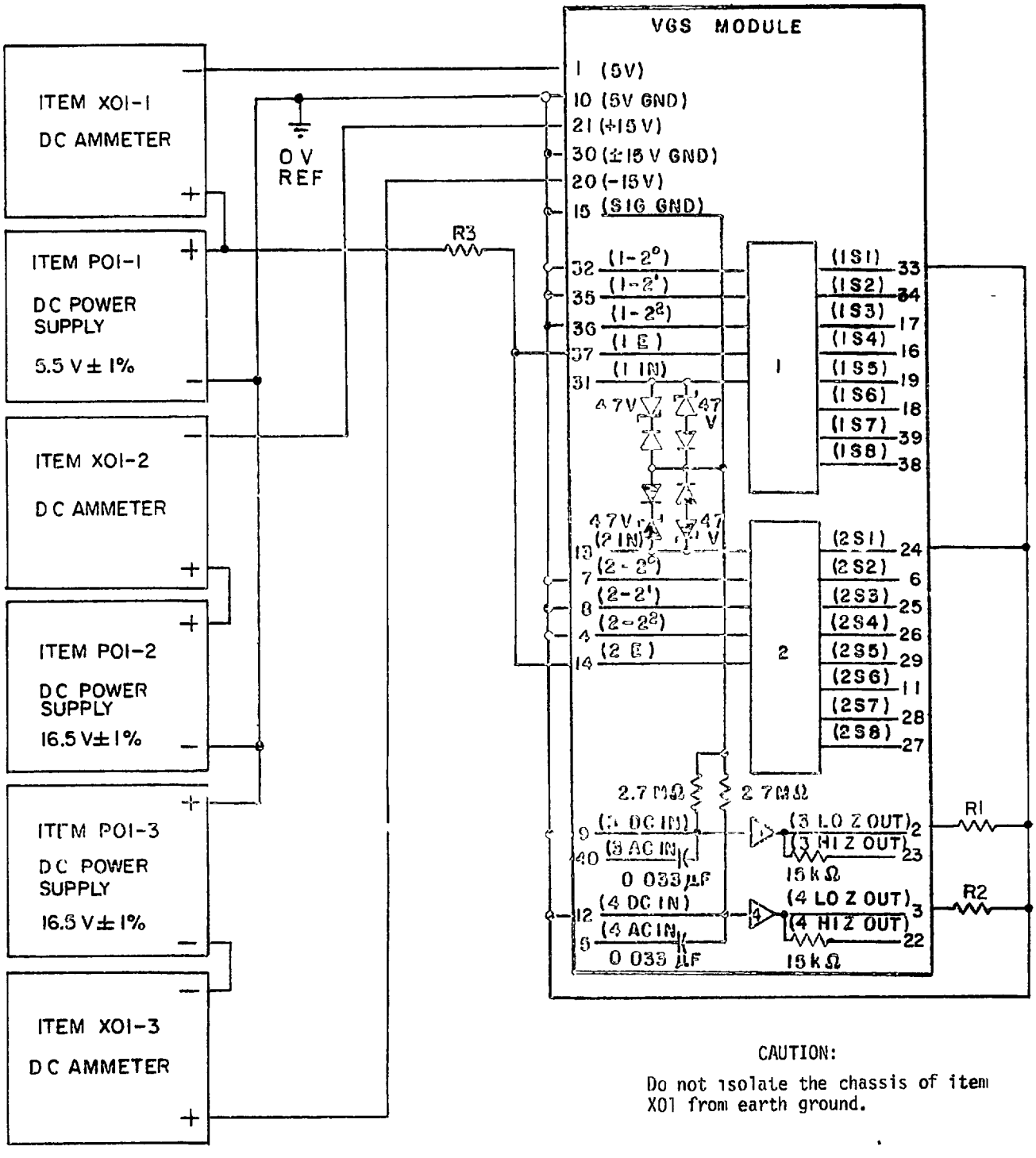
WARNING

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish of the module frame with a sharply pointed probe connected to the negative terminal.

*FIGURE 7. Isolation test circuit for table V.

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTES:
1. R1 = R2 = 2 kΩ ±1%, 1/4 W.
2. R3 = 1 kΩ ±1%, 1/4 W.

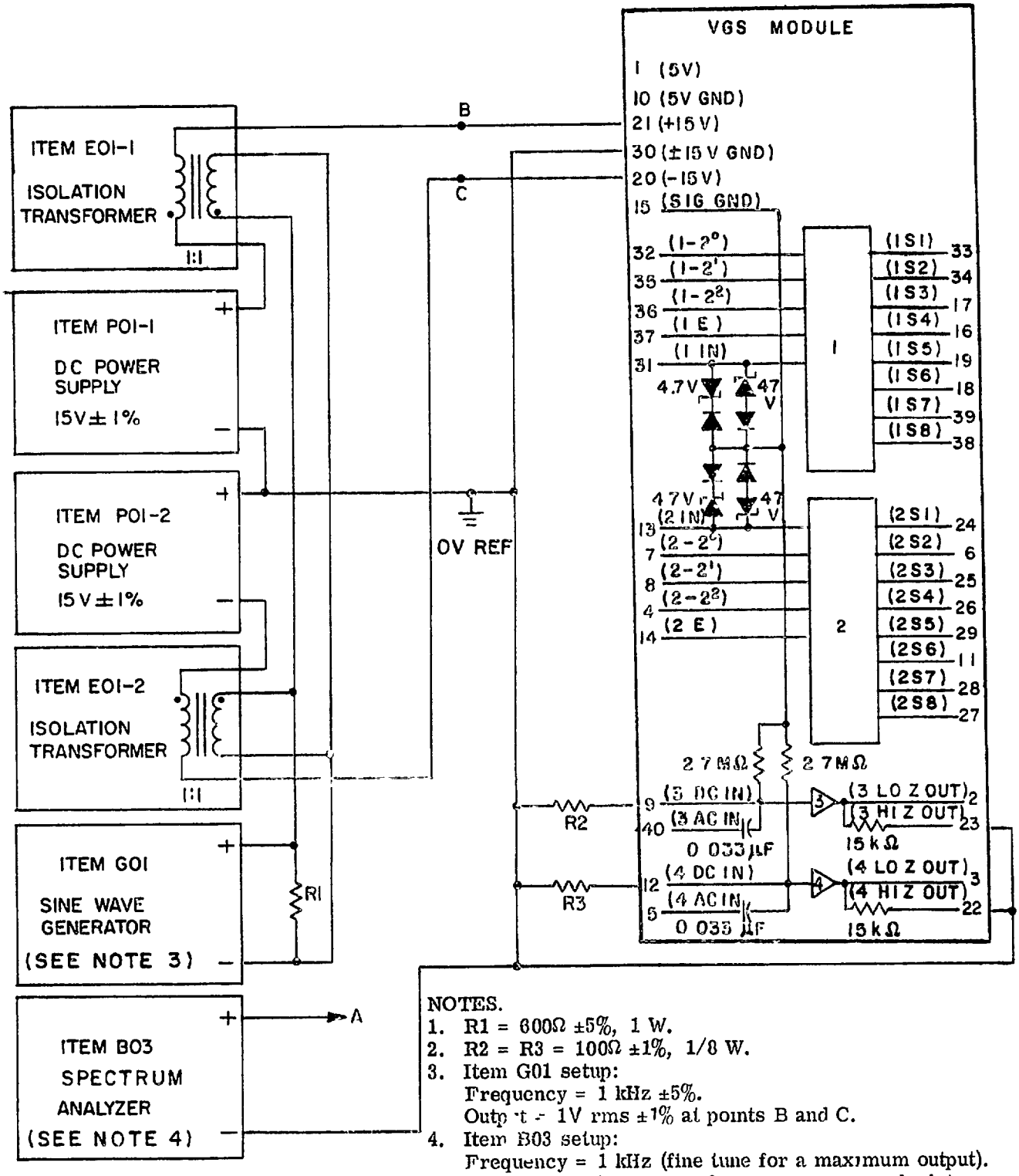
*FIGURE 8. Power supply current test circuit for table VI.

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TABLE VII. Supply voltage rejection ratio check test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	3HI Z OUT Supply voltage rejection ratio check	2	0	316	µV	9
E2	4HI Z OUT Supply voltage rejection ratio check	3	0	316	µV	9

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NOTES.
1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 100Ω ±1%, 1/8 W.
3. Item G01 setup:
   Frequency = 1 kHz ±5%.
   Output = 1V rms ±1% at points B and C.
4. Item B03 setup:
   Frequency = 1 kHz (fine tune for a maximum output).
   Bandwidth = 1 Hz or perform conversion calculation from wider bandwidth as specified by wave analyzer manufacturer.
5. This test verifies a supply voltage rejection ratio of -70 dB minimum.

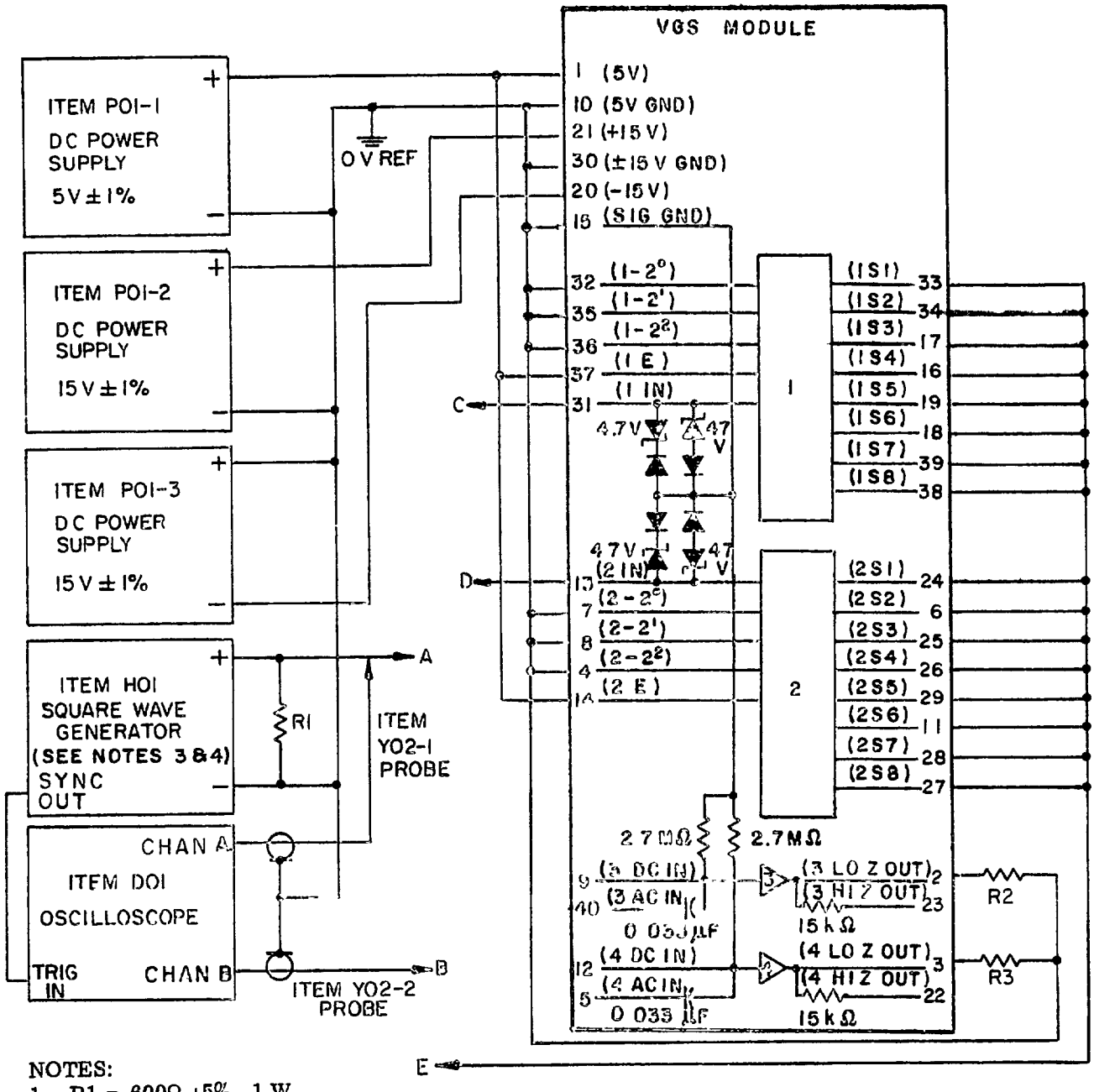
*FIGURE 9. Supply voltage rejection ratio check test circuit for table VII.

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*TABLE VIII. Power supply transient amplitude and slew rate check test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	Power supply transient amplitude	E	21	9	12	0	300	mV	10,11
F2	"	"	20	"	"	"	"	"	"
F3	"	"	1	"	"	"	"	"	"
G1	Slew rate check, negative to positive	9	2	Open	Open	10	600	ns	10,12
G2	Slew rate check, positive to negative	9	2	"	"	"	"	"	"
G3	Slew rate check, negative to positive	12	3	"	"	"	"	"	"
G4	Slew rate check, positive to negative	12	3	"	"	"	"	"	"

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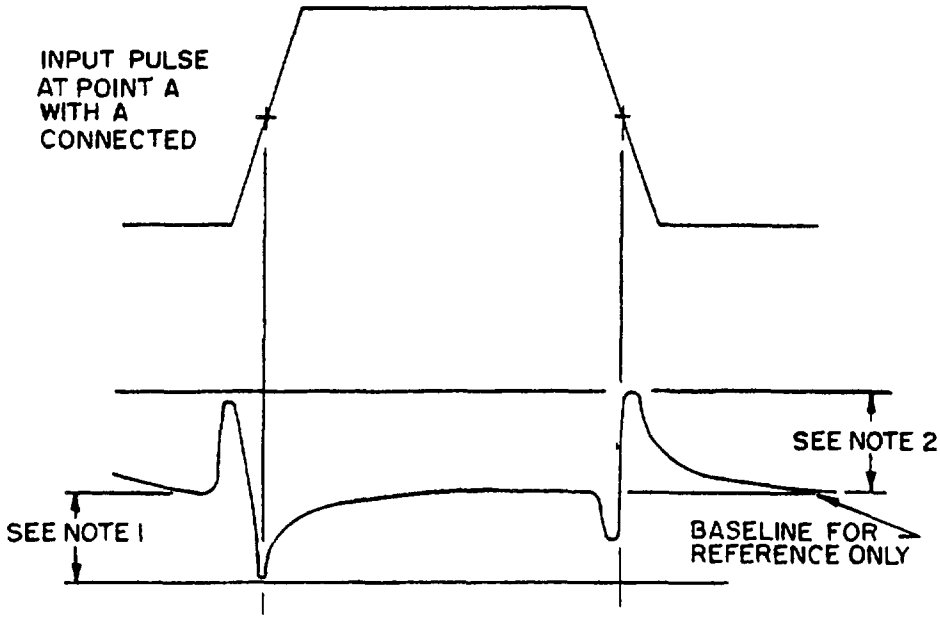


NOTES:

1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 10 kΩ ±1%, 1/4 W.
3. Item H01 setup for test condition F:
   Amplitude = 8.0 ±0.5 V_{p-p}
   Frequency = 1 kHz ±10%.
4. Item H01 setup for test condition G:
   Amplitude = 20 V_{p-p} ±10%.
   Frequency = 10 kHz ±10%.
5. Turn on items P01-1, P01-2, and P01-3 prior to applying signal from item H01 to the module. Turn off item H01 before turning off P01-1, P01-2, and P01-3. Module damage will result if the sequence is not followed.

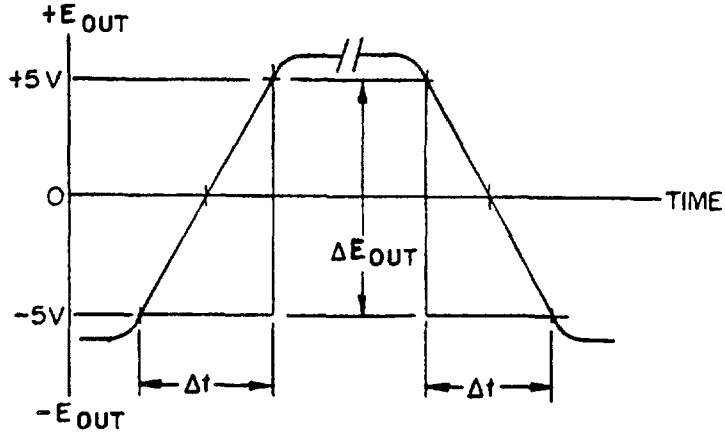
*FIGURE 10. Power supply transient amplitude and slew rate check test circuit for table VIII.

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NOTES:
1. For test conditions F1 and F3, measurement is made from the baseline to the most negative peak.
2. For test condition F2, measurement is made from the baseline to the most positive peak.

FIGURE 11. Power supply transient amplitude test circuit waveform for table VIII.



$$\text{Slew rate} = \frac{\Delta E_{OUT}}{\Delta t} = \frac{10}{\Delta t}$$

$$\Delta t = \frac{10\ \text{V}}{167\ \text{V}/\mu\text{s}} = 0.60\ \mu\text{s maximum}$$

*FIGURE 12. Slew rate check test circuit waveform for table VIII.

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TABLE IX. Input current (low-level and high-level) test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	1-E	Input current, low level	Open	37	32,35,36	Open	0.1	1.2	mA	13
H2	1-E	Input current, high level	37	Open	Open	32,35,36	0	-350	µA	"
H3	$1\text{-}2^0$	Input current, low level	Open	32	35,36,37	Open	0.1	1.2	mA	"
H4	$1\text{-}2^0$	Input current, high level	32	Open	Open	35,36,37	0	-350	µA	"
H5	$1\text{-}2^1$	Input current, low level	Open	35	32,36,37	Open	0.1	1.2	mA	"
H6	$1\text{-}2^1$	Input current, high level	35	Open	Open	32,36,37	0	-350	µA	"
H7	$1\text{-}2^2$	Input current, low level	Open	36	32,35,37	Open	0.1	1.2	mA	"
H8	$1\text{-}2^2$	Input current, high level	36	Open	Open	32,35,37	0	-350	µA	"
H9	2-E	Input current, low level	Open	14	4,7,8	Open	0.1	1.2	mA	"
H10	2-E	Input current, high level	14	Open	Open	4,7,8	0	-350	µA	"
H11	$2\text{-}2^0$	Input current, low level	Open	7	4,8,14	Open	0.1	1.2	mA	"
H12	$2\text{-}2^0$	Input current, high level	7	Open	Open	4,8,14	0	-350	µA	"
H13	$2\text{-}2^1$	Input current, low level	Open	8	4,7,14	Open	0.1	1.2	mA	"
H14	$2\text{-}2^1$	Input current, high level	8	Open	Open	4,7,14	0	-350	µA	"
H15	$2\text{-}2^2$	Input current, low level	Open	4	7,8,14	Open	0.1	1.2	mA	"
H16	$2\text{-}2^2$	Input current, high level	4	Open	Open	7,8,14	0	-350	µA	"

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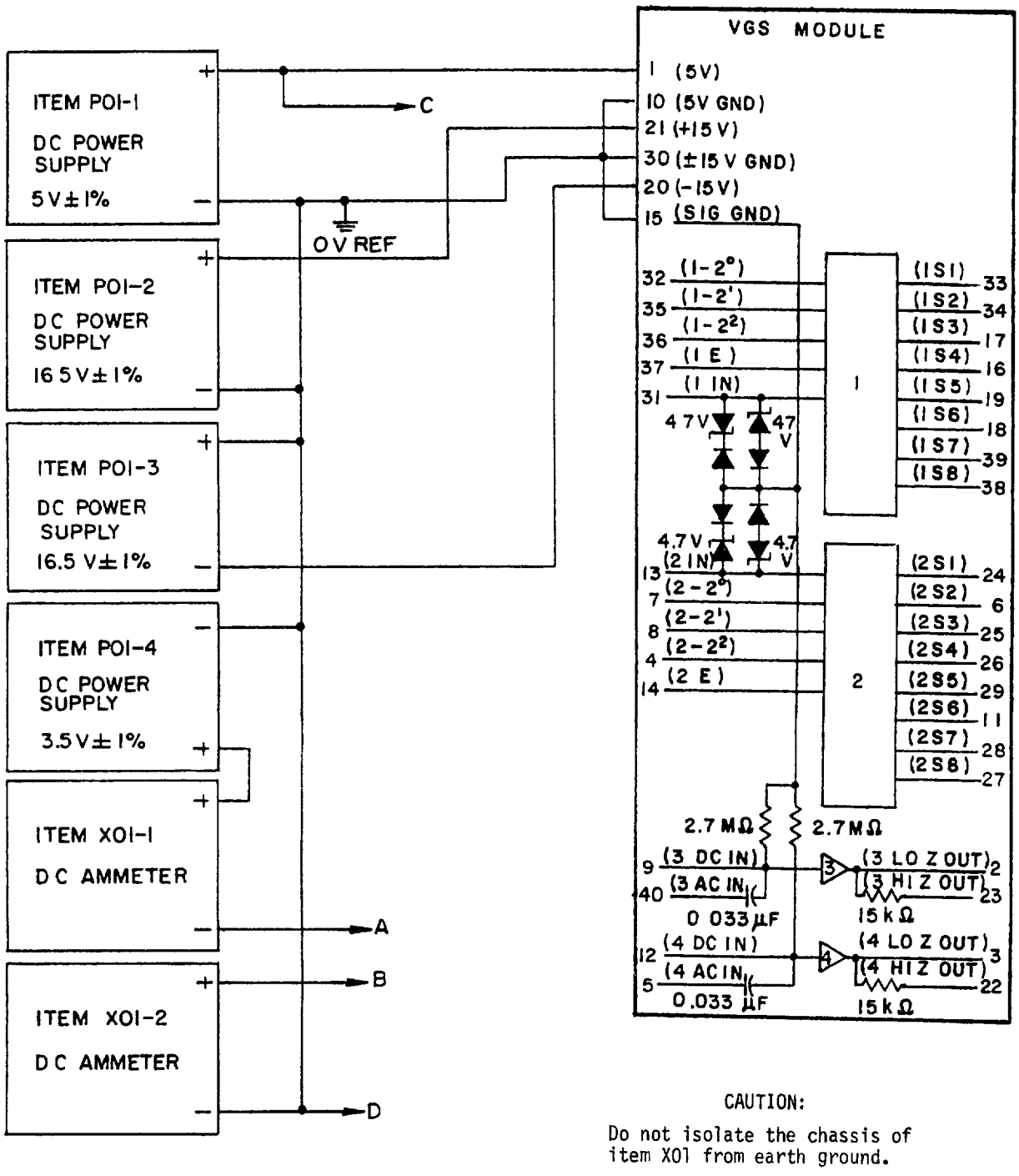


FIGURE 13. Input current (low-level and high-level) test circuit for table IX.

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TABLE X. ON resistance check test.

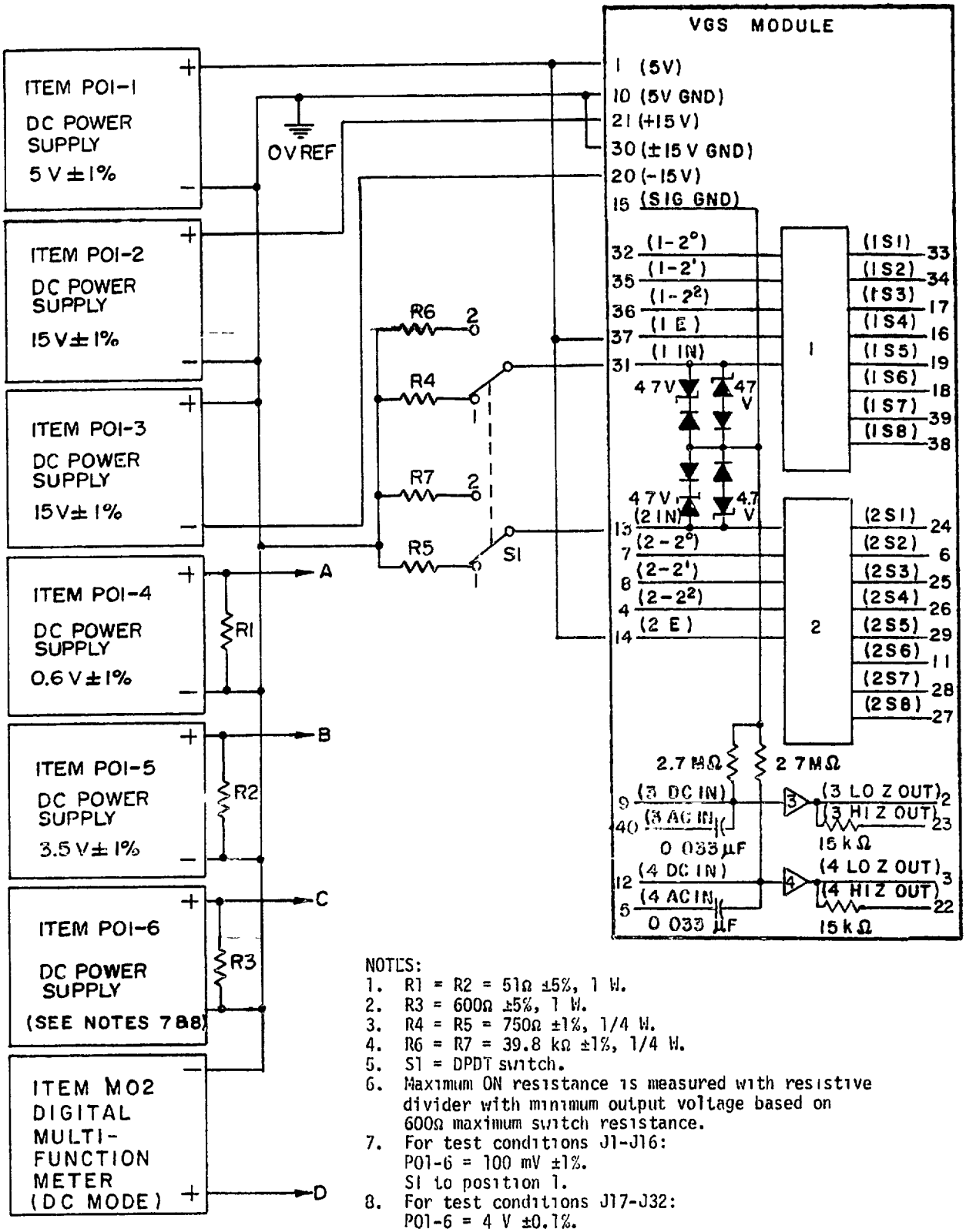
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J1	1S1 ON resistance check, V_I = 100 mV	32,35,36	Open	33	31	75	100	mV	14
J2	1S2 "	35,36	32	34	"	"	"	"	"
J3	1S3 "	32,36	35	17	"	"	"	"	"
J4	1S4 "	36	32,35	16	"	"	"	"	"
J5	1S5 "	32,35	36	19	"	"	"	"	"
J6	1S6 "	35	32,36	18	"	"	"	"	"
J7	1S7 "	32	35,36	39	"	"	"	"	"
J8	1S8 "	Oper	32,35,36	38	"	"	"	"	"
J9	2S1 "	4,7,8	Open	24	13	"	"	"	"
J10	2S2 "	4,8	7	6	"	"	"	"	"
J11	2S3 "	4,7	8	25	"	"	"	"	"
J12	2S4 "	4	7,8	26	"	"	"	"	"
J13	2S5 "	7,8	4	29	"	"	"	"	"
J14	2S6 "	8	4,7	11	"	"	"	"	"
J15	2S7 "	7	4,8	28	"	"	"	"	"
J16	2S8 "	Open	4,7,8	27	"	"	"	"	"
J17	1S1 ON resistance check, V_I = 4 V	32,35,36	Open	33	31	3.94	4.04	V	"
J18	1S2 "	35,36	32	34	"	"	"	"	"
J19	1S3 "	32,36	35	17	"	"	"	"	"
J20	1S4 "	36	32,35	16	"	"	"	"	"
J21	1S5 "	32,35	36	19	"	"	"	"	"

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TABLE X. ON resistance check test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
J22	1S6 ON resistance check, V_I = 4 V	35	32,36	18	31	3.94	4.04	V	14
J23	1S7 "	32	35,36	39	"	"	"	"	"
J24	1S8 "	Open	32,35,36	38	"	"	"	"	"
J25	2S1 "	4,7,8	Open	24	13	"	"	"	"
J26	2S2 "	4,8	7	6	"	"	"	"	"
J27	2S3 "	4,7	8	25	"	"	"	"	"
J28	2S4 "	4	7,8	26	"	"	"	"	"
J29	2S5 "	7,8	4	29	"	"	"	"	"
J30	2S6 "	8	4,7	11	"	"	"	"	"
J31	2S7 "	7	4,8	28	"	"	"	"	"
J32	2S8 "	Open	4,7,8	27	"	"	"	"	"

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NOTES:
1. R1 = R2 = 51Ω ±5%, 1 W.
2. R3 = 600Ω ±5%, 1 W.
3. R4 = R5 = 750Ω ±1%, 1/4 W.
4. R6 = R7 = 39.8 kΩ ±1%, 1/4 W.
5. S1 = DPDT switch.
6. Maximum ON resistance is measured with resistive divider with minimum output voltage based on 600Ω maximum switch resistance.
7. For test conditions J1-J16:
   POI-6 = 100 mV ±1%.
   S1 to position 1.
8. For test conditions J17-J32:
   POI-6 = 4 V ±0.1%.
   S1 to position 2.
9. Wait one minute prior to taking measurements.

*FIGURE 14. ON resistance check test circuit for table X.

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TABLE VI. Large signal voltage response and distortion test.

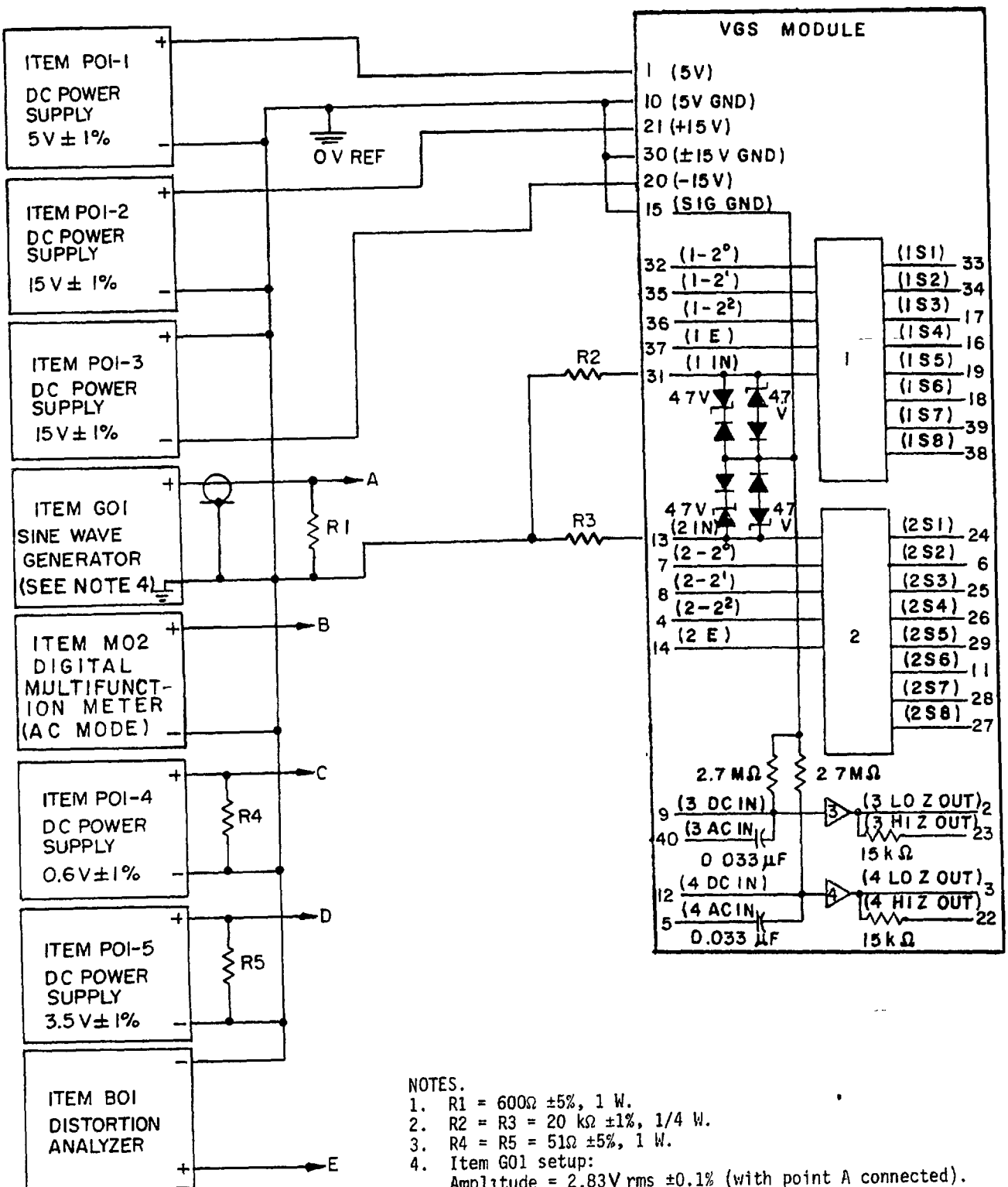
Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
K1	1S1 Large signal voltage response	33	31	32,35,36	37	Open	2.75	2.83	V rms	15
K2	1S2 "	34	"	35,36	32,37	"	"	"	"	"
K3	1S3 "	17	"	32,36	35,37	"	"	"	"	"
K4	1S4 "	16	"	36	32,35,37	"	"	"	"	"
K5	1S5 "	19	"	32,35	36,37	"	"	"	"	"
K6	1S6 "	18	"	35	32,36,37	"	"	"	"	"
K7	1S7 "	39	"	32	35,36,37	"	"	"	"	"
K8	1S8 "	38	"	Open	32,35,36,37	"	"	"	"	"
K9	2S1 "	24	13	4,7,8	14	"	"	"	"	"
K10	2S2 "	6	"	4,8	7,14	"	"	"	"	"
K11	2S3 "	25	"	4,7	8,14	"	"	"	"	"
K12	2S4 "	26	"	4	7,8,14	"	"	"	"	"
K13	2S5 "	29	"	7,8	4,14	"	"	"	"	"
K14	2S6 "	11	"	8	4,7,14	"	"	"	"	"
K15	2S7 "	28	"	7	4,8,14	"	"	"	"	"
K16	2S8 "	27	"	Open	4,7,8,14	"	"	"	"	"
K17	1S1 Distortion	33	Open	32,35,36	37	31	0	1	%	"
K18	1S2 "	34	"	35,36	32,37	"	"	"	"	"
K19	1S3 "	17	"	32,36	35,37	"	"	"	"	"
K20	1S4 "	16	"	36	32,35,37	"	"	"	"	"
K21	1S5 "	19	"	32,35	36,37	"	"	"	"	"

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TABLE XI. Large signal voltage response and distortion test - Continued.

Test condition	Parameter		Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
K22	1S6	Distortion	18	Open	35	32,36,37	31	0	1	%	15
K23	1S7	"	39	"	32	35,36,37	"	"	"	"	"
K24	1S8	"	38		Open	32,35,36,37	"	"	"	"	"
K25	2S1	"	24	"	4,7,8	14	13	"	"	"	"
K26	2S2	"	6	"	4,8	7,14	"	"	"	"	"
K27	2S3	"	25	"	4,7	8,14	"	"	"	"	"
K28	2S4	"	26	"	4	7,8,14	"	"	"	"	"
K29	2S5	"	29	"	7,8	4,14	"	"	"	"	"
K30	2S6	"	11	"	8	4,7,14	"	"	"	"	"
K31	2S7	"	28	"	7	4,8,14	"	"	"	"	"
K32	2S8	"	27	"	Open	4,7,8,14	"	"	"	"	"

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NOTES.
1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 20 kΩ ±1%, 1/4 W.
3. R4 = R5 = 51Ω ±5%, 1 W.
4. Item GO1 setup:
   Amplitude = 2.83V rms ±0.1% (with point A connected).
   Frequency = 20 kHz ±1%.
5. These tests verify threshold control voltages.
6. Subtract input distortion from output distortion before comparing to table XI limits.

*FIGURE 15. Large signal voltage response and distortion test circuit for table XI.

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*TABLE XII. Large signal channel-to-channel attenuation test.

Test condition	Parameter		Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	D to pin	Min	Max		
L1	1S1	Large signal channel-to-channel attenuation, enable off	33	31	32,35,36,37	Open	0	5	mV rms	16
L2	1S1	Large signal channel-to-channel attenuation, enable on	33	"	35,36	32,37	"	"	"	"
L3	1S2	"	34	"	32,36	35,37	"	"	"	"
L4	1S3	"	17	"	36	32,35,37	"	"	"	"
L5	1S4	"	16	"	32,35	36,37	"	"	"	"
L6	1S5	"	19	"	35	32,36,37	"	"	"	"
L7	1S6	"	18	"	32	35,36,37	"	"	"	"
L8	1S7	"	39	"	Open	32,35, 36,37	"	"	"	"
L9	1S8	"	38	"	32,35,36	37	"	"	"	"
L10	2S1	Large signal channel-to-channel attenuation, enable off	24	13	4,7,8,14	Open	"	"	"	"
L11	2S1	Large signal channel-to-channel attenuation, enable on	24	"	4,8	7,14	"	"	"	"
L12	2S2	"	6	"	4,7	8,14	"	"	"	"
L13	2S3	"	25	"	4	7,8,14	"	"	"	"
L14	2S4	"	26	"	7,8	4,14	"	"	"	"
L15	2S5	"	29	"	8	4,7,14	"	"	"	"
L16	2S6	"	11	"	7	4,8,14	"	"	"	"
L17	2S7	"	28	"	Open	4,7,8,14	"	"	"	"
L18	2S8	"	27	"	4,7,8	14	"	"	"	"

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TABLE XIII. Crosstalk check test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
M1	Crosstalk check, circuit 2 to circuit 1	16,17,18,19,33, 34,38,39	13	32,35,36	37	0	1.8	mV rms	16
M2	"	"	"	35,36	32,37	"	"	"	"
M3	"	"	"	32,36	35,37	"	"	"	"
M4	"	"	"	36	32,35,37	"	"	"	"
M5	"	"	"	32,35	36,37	"	"	"	"
M6	"	"	"	35	32,36,37	"	"	"	"
M7	"	"	"	32	35,36,37	"	"	"	"
M8	"	"	"	Open	32,35, 36,37	"	"	"	"
M9	Crosstalk check, circuit 1 to circuit 2	6,11,24,25,26, 27,28,29	31	4,7,8	14	"	895	µV rms	"
M10	"	"	"	4,8	7,14	"	"	"	"
M11	"	"	"	4,7	8,14	"	"	"	"
M12	"	"	"	4	7,8,14	"	"	"	"
M13	"	"	"	7,8	4,14	"	"	"	"
M14	"	"	"	8	4,7,14	"	"	"	"
M15	"	"	"	7	4,8,14	"	"	"	"
M16	"	"	"	Open	4,7,8,14	"	"	"	"

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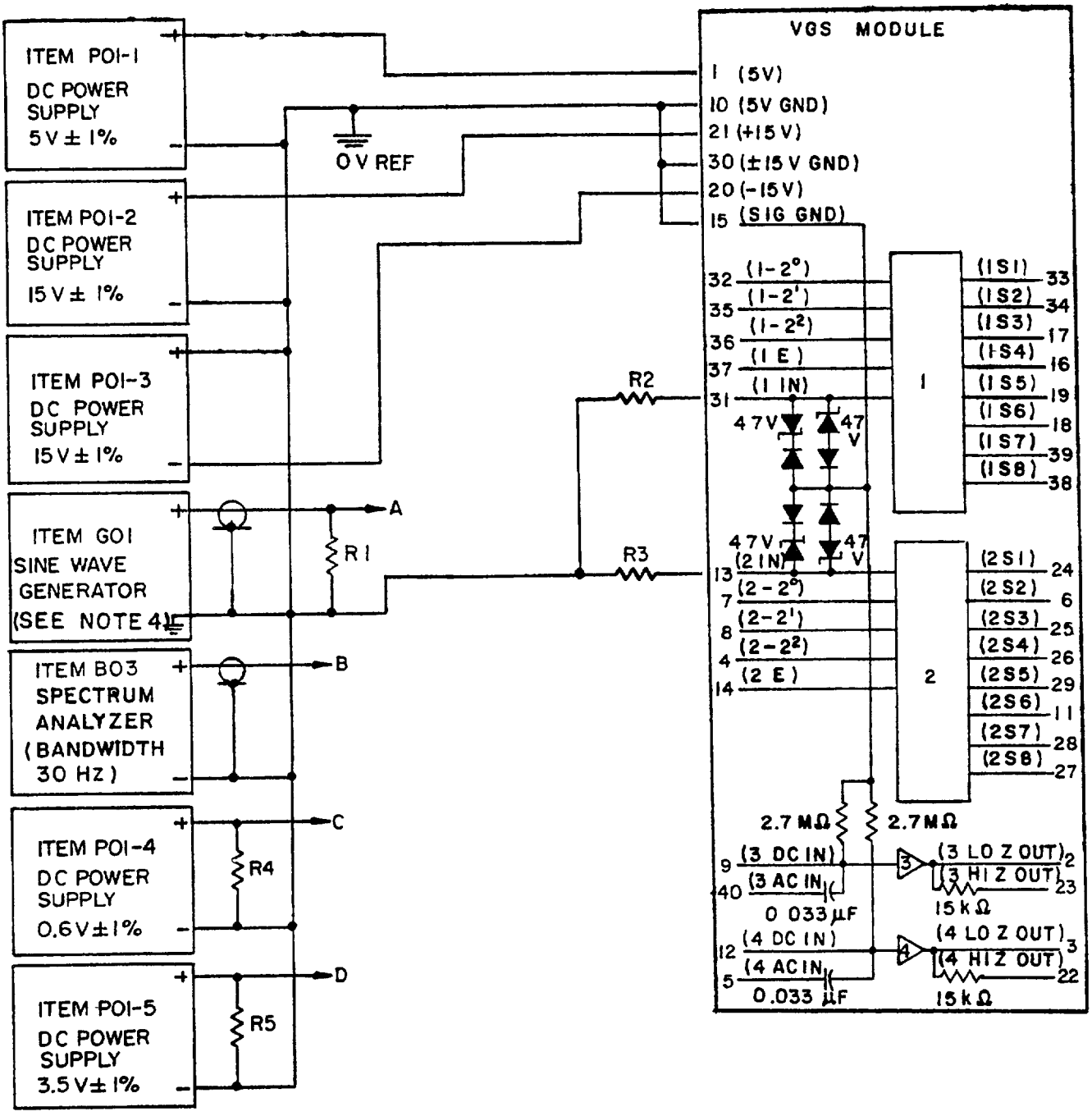


*FIGURE 16. Large signal channel-to-channel attenuation and crosstalk check test circuit for tables XII and XIII.

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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = R3 = 15 kΩ ±1%, 1/4 W.
3. R4 = R5 = 51Ω ±5%, 1 W.
4. Item G01 setup:
   Amplitude = 2.83V rms ±0.1% (with point A connected).
   Frequency = 15 kHz ±1% for test condition L and 10 kHz ±1% for test condition M.
5. These tests verify threshold control voltages.
6. Test condition L verifies channel off isolation of 55 dB.
7. Test condition M verifies circuit-to-circuit isolation of 64 dB.
8. For test conditions L and M, connect the mounting plate of the module connector receptacle to 0 V REF (ground).
9. For test condition M connect each unused demultiplexer output (S1 through S8) to ground through a 10 kΩ ±1%, 1/4 W, resistor (8 required).
10. The test fixture must include extra shielding to assure that all voltage being measured by item B03 actually comes from the module. The wire and/or printed circuit board conductor from item G01 (plus terminal) to and including the VGS module mating female connector must be totally shielded. The wire and/or printed circuit board conductor from item B03 (plus terminal) to and including the VGS module mating connector must be totally shielded.

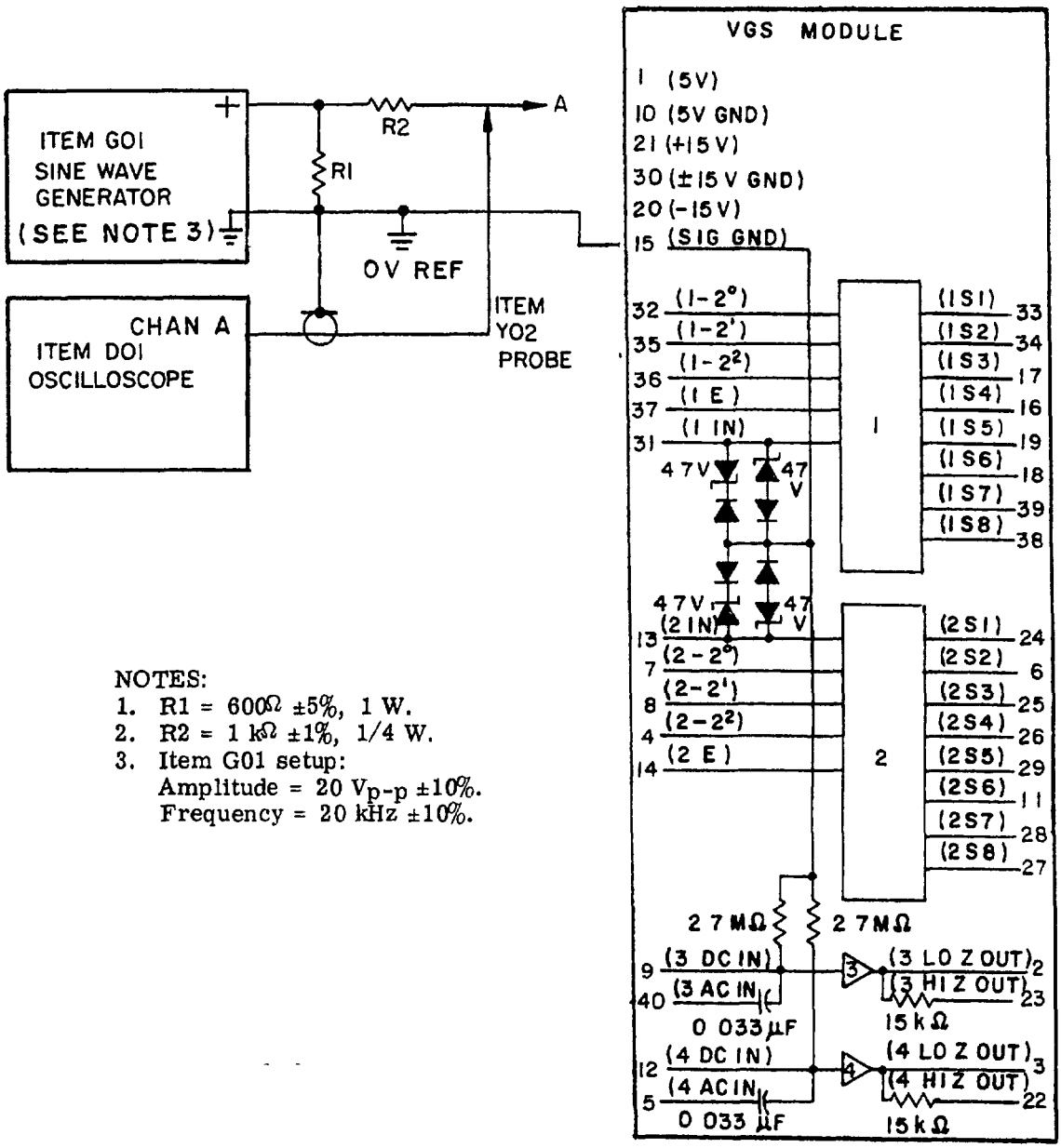
FIGURE 16 Large signal channel-to-channel attenuation and crosstalk check test circuit for tables XII and XIII - Continued.

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TABLE XIV. Input clamp diode voltage test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
N1	2IN Input clamp diode voltage	13	4.4	5.7	V	17,18
N2	1IN "	31	4.4	5.7	"	"
N3	2IN "	13	-4.4	-5.7	"	"
N4	1IN "	31	-4.4	-5.7	"	"

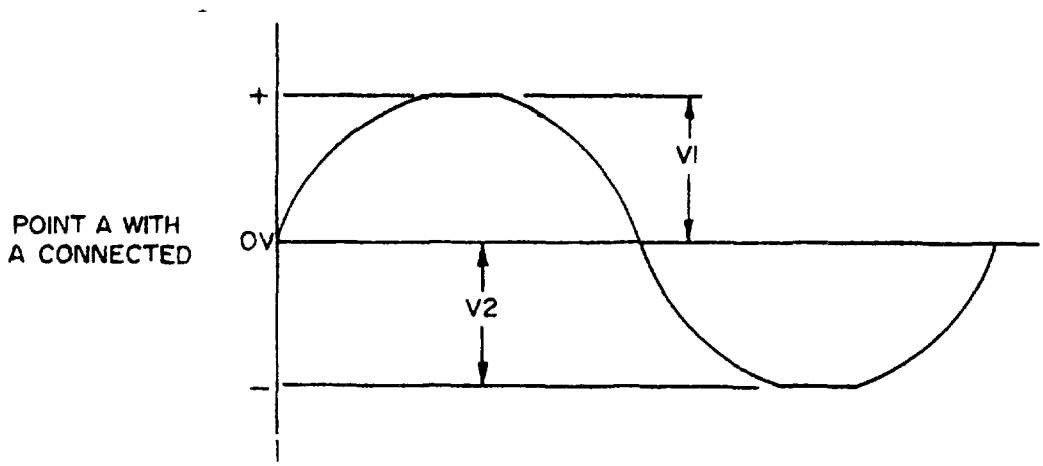
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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2 = 1 kΩ ±1%, 1/4 W.
3. Item G01 setup:
   Amplitude = 20 V_{p-p} ±10%.
   Frequency = 20 kHz ±10%.

*FIGURE 17. Input clamp diode voltage test circuit for table XIV.

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NOTES.
1. For test conditions N1 and N2, V1 shall meet the limits specified in table XIV.
2. For test conditions N3 and N4, V2 shall meet the limits specified in table XIV.

*FIGURE 18. Input clamp diode voltage test circuit waveform for table XIV.

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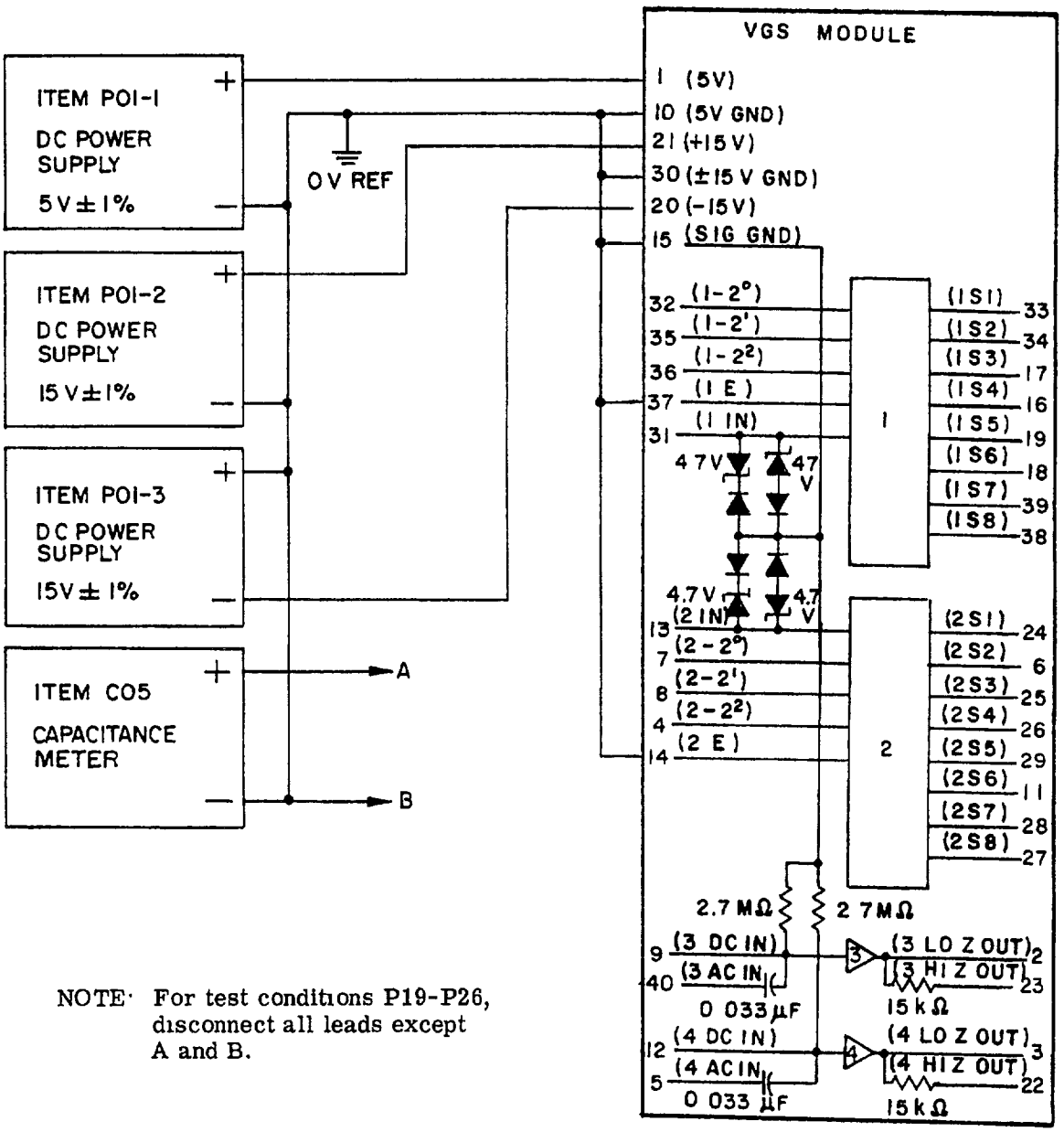
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
P1	1IN	Capacitance	31	16,17,18,19, 33,34,38,39	5	100	pF	19
P2	2IN	"	13	6,11,24,25, 26,27,28,29	"	100	"	"
P3	1S1	"	33	31	"	25	"	"
P4	1S2	"	34	"	"	"	"	"
P5	1S3	"	17	"	"	"	"	"
P6	1S4	"	16	"	"	"	"	"
P7	1S5	"	19	"	"	"	"	"
P8	1S6	"	18	"	"	"	"	"
P9	1S7	"	39	"	"	"	"	"
P10	1S8	"	38	"	"	"	"	"
P11	2S1	"	24	13	"	"	"	"
P12	2S2	"	6	"	"	"	"	"
P13	2S3	"	25	"	"	"	"	"
P14	2S4	"	26	"	"	"	"	"
P15	2S5	"	29	"	"	"	"	"
P16	2S6	"	11	"	"	"	"	"
P17	2S7	"	28	"	"	"	"	"
P18	2S8	"	27	"	"	"	"	"
P19	1E	"	37	20	"	"	"	"
P20	$1\text{-}2^0$	"	32	"	"	"	"	"
P21	$1\text{-}2^1$	"	35	"	"	"	"	"

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TABLE XV. Capacitance test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
P22	1-2^2	Capacitance	36	20	5	25	pF	19
P23	2E	"	14	"	"	"	"	"
P24	2-2^0	"	7	"	"	"	"	"
P25	2-2^1	"	8	"	"	"	"	"
P26	2-2^2	"	4	"	"	"	"	"

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NOTE: For test conditions P19-P26, disconnect all leads except A and B.

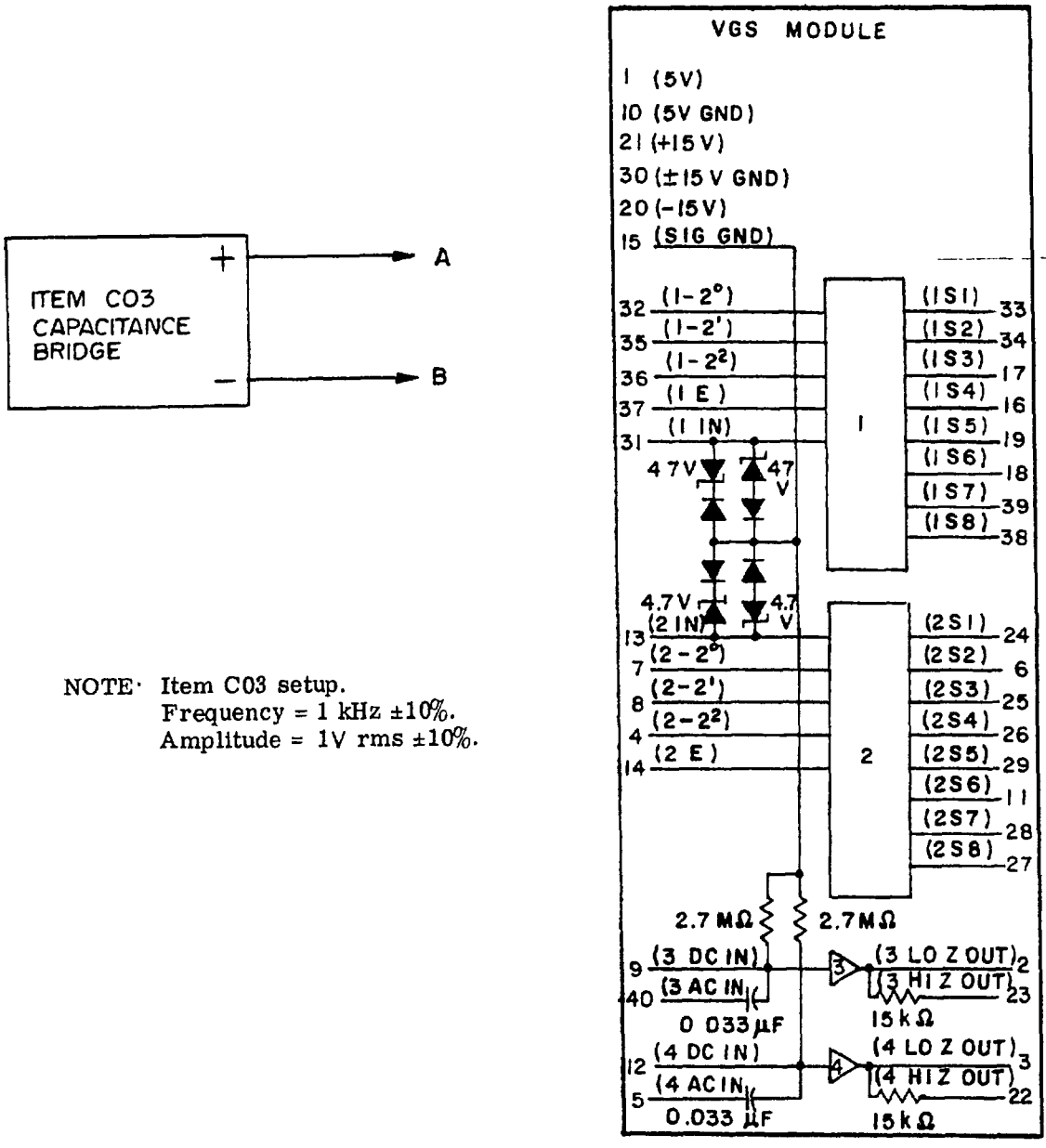
*FIGURE 19. Capacitance test curcuit for table XV.

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TABLE XVI. Buffer coupling capacitance and buffer coupling dissipation factor test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Q1	Buffer coupling capacitance	40	9	0.026	0.040	µF	20
Q2	Buffer coupling capacitance	5	12	0.026	0.040	µF	"
Q3	Buffer coupling dissipation factor	40	9	0	5	%	"
Q4	Buffer coupling dissipation factor	5	12	0	5	%	"

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NOTE: Item C03 setup.
Frequency = 1 kHz ±10%.
Amplitude = 1V rms ±10%.

*FIGURE 20. Buffer coupling capacitance and buffer coupling dissipation factor test circuit for table XVI.

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TABLE XVII. Leakage current test.

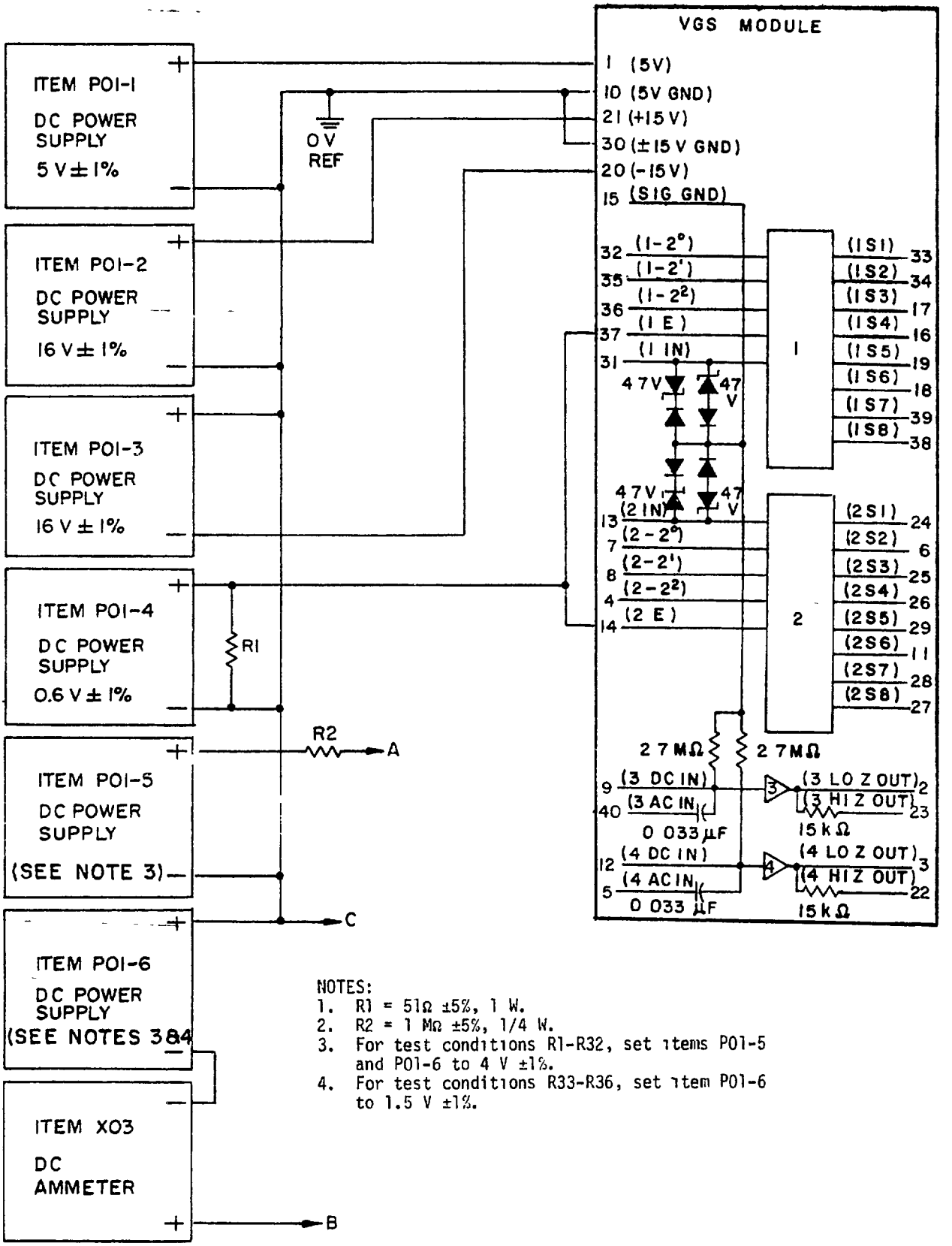
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R1	1S1	Leakage current, output	31	33	Open	-5	100	nA	21
R2	1S2	"	"	34	"	"	"	"	"
R3	1S3	"	"	17	"	"	"	"	"
R4	1S4	"	"	16	"	"	"	"	"
R5	1S5	"	"	19	"	"	"	"	"
R6	1S6	"	"	18	"	"	"	"	"
R7	1S7	"	"	39	"	"	"	"	"
R8	1S8	"	"	38	"	"	"	"	"
R9	2S1	"	13	24	"	"	"	"	"
R10	2S2	"	"	6	"	"	"	"	"
R11	2S3	"	"	25	"	"	"	"	"
R12	2S4	"	"	26	"	"	"	"	"
R13	2S5	"	"	29	"	"	"	"	"
R14	2S6	"	"	11	"	"	"	"	"
R15	2S7	"	"	28	"	"	"	"	"
R16	2S8	"	"	27	"	"	"	"	"
R17	1S1	Leakage current, reverse	33	31	"	"	"	"	"
R18	1S2	"	34	"	"	"	"	"	"
R19	1S3	"	17	"	"	"	"	"	"
R20	1S4	"	16	"	"	"	"	"	"
R21	1S5	"	19	"	"	"	"	"	"

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TABLE XVII. Leakage current test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R22	1S6 Leakage current, reverse	18	31	Open	-5	100	nA	21
R23	1S7 "	39	"	"	"	"	"	"
R24	1S8 "	38	"	"	"	"	"	"
R25	2S1 "	24	13	"	"	"	"	"
R26	2S2 "	6	"	"	"	"	"	"
R27	2S3 "	25	"	"	"	"	"	"
R28	2S4 "	26	"	"	"	"	"	"
R29	2S5 "	29	"	"	"	"	"	"
R30	2S6 "	11	"	"	"	"	"	"
R31	2S7 "	28	"	"	"	"	"	"
R32	2S8 "	27	"	"	"	"	"	"
R33	Leakage current, diodes	Open	15	31	0	10	µA	"
R34	"	"	31	15	"	"	"	"
R35	"	"	15	13	"	"	"	"
R36	"	"	13	15	"	"	"	"

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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 1 MΩ ±5%, 1/4 W.
3. For test conditions R1-R32, set items P01-5 and P01-6 to 4 V ±1%.
4. For test conditions R33-R36, set item P01-6 to 1.5 V ±1%.

FIGURE 21. Leakage current test circuit for table XVII.

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TABLE XVIII. Switching time test.

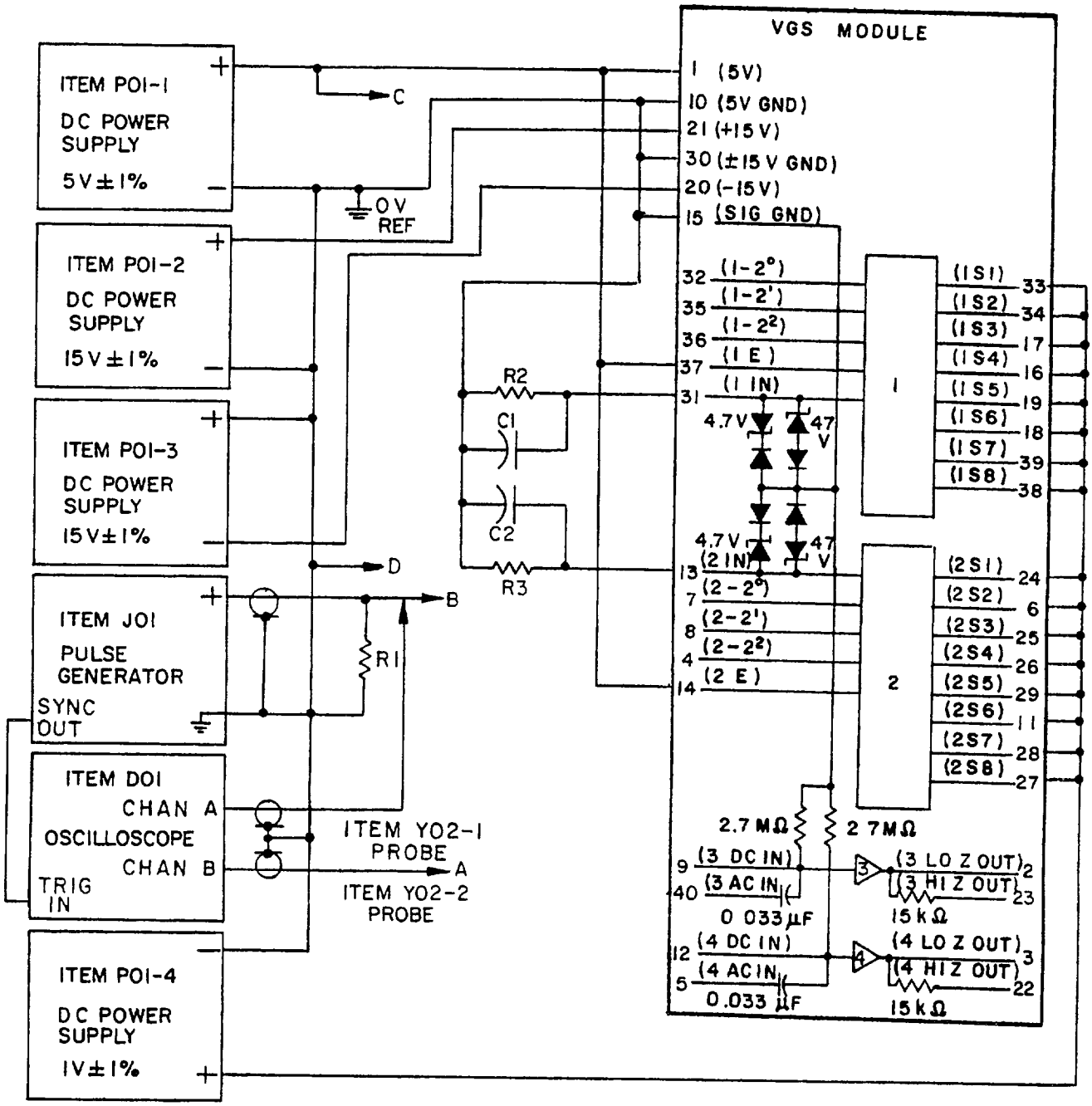
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1S1 Switching time, t_{ON}	31	32	Open	35,36	0.1	3.0	µs	22,23
S2	1S2 "	"	"	Open	35,36	"	"	"	"
S3	1S3 "	"	"	35	36	"	"	"	"
S4	1S4 "	"	"	35	36	"	"	"	"
S5	1S5 "	"	"	36	35	"	"	"	"
S6	1S6 "	"	"	36	35	"	"	"	"
S7	1S7 "	"	"	35,36	Open	"	"	"	"
S8	1S8 "	"	"	35,36	Open	"	"	"	"
S9	2S1 "	13	7	Open	4,8	"	"	"	"
S10	2S2 "	"	"	Open	4,8	"	"	"	"
S11	2S3 "	"	"	8	4	"	"	"	"
S12	2S4 "	"	"	8	4	"	"	"	"
S13	2S5 "	"	"	4	8	"	"	"	"
S14	2S6 "	"	"	4	8	"	"	"	"
S15	2S7 "	"	"	4,8	Open	"	"	"	"
S16	2S8 "	"	"	4,8	Open	"	"	"	"
S17	1S1 Switching time, t_{OFF}	31	32	Open	35,36	"	2.0	"	"
S18	1S2 "	"	"	Open	35,36	"	"	"	"
S19	1S3 "	"	"	35	36	"	"	"	"
S20	1S4 "	"	"	35	36	"	"	"	"
S21	1S5 "	"	"	36	35	"	"	"	"

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TABLE XVIII. Switching time test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S22	1S6 Switching time, t_{OFF}	31	32	36	35	0.1	2.0	µs	22,23
S23	1S7 "	"	"	35,36	Open	"	"	"	"
S24	1S8 "	"	"	35,36	Open	"	"	"	"
S25	2S1 "	13	7	Open	4,8	"	"	"	"
S26	2S2 "	"	"	Open	4,8	"	"	"	"
S27	2S3 "	"	"	8	4	"	"	"	"
S28	2S4 "	"	"	8	4	"	"	"	"
S29	2S5 "	"	"	4	8	"	"	"	"
S30	2S6 "	"	"	4	8	"	"	"	"
S31	2S7 "	"	"	4,8	Open	"	"	"	"
S32	2S8 "	"	"	4,8	"	"	"	"	"
S33	Switching time, t_{OPEN}	31	32,35,36	Open	"	0.05	3.00	"	"
S34	Switching time, t_{OPEN}	13	4,7,8	Open	"	0.05	3.00	"	"

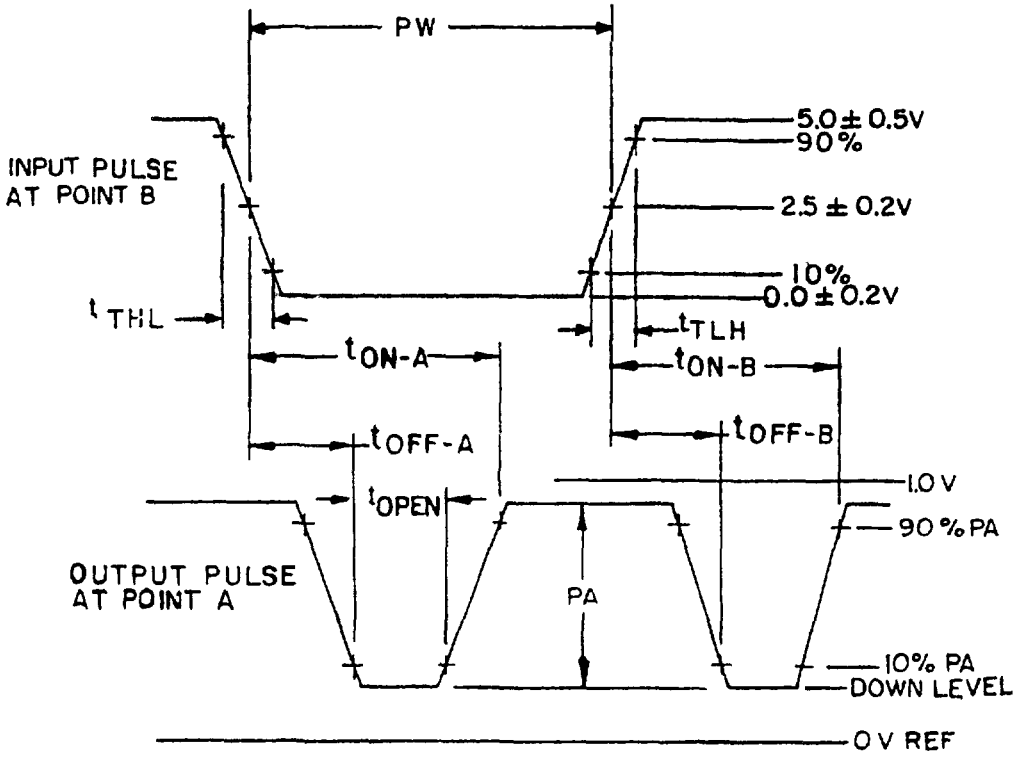
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = R3 = 1 kΩ ±1%, 1/4 W.
3. C1 = C2 = 30 pF ±5% (including probe capacitance

* FIGURE 22. Switching time test circuit for table XVIII.

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NOTES:
1. Item J01 setup:
   PW = 5 µs.
   $t_{TLH} = t_{THL} = 20 \pm 2$ ns.
2. All voltage levels are referenced to ground.
3. Measure t_{ON-A} for test conditions: S1, S3, S5, S7, S9, S11, S13, S15.
4. Measure t_{ON-B} for test conditions: S2, S4, S6, S8, S10, S12, S14, S16.
5. Measure t_{OFF-A} for test conditions: S18, S20, S22, S24, S26, S28, S30, S32.
6. Measure t_{OFF-B} for test conditions: S17, S19, S21, S23, S25, S29, S31.
7. Before making these measurements, verify that the down level is 100 mV maximum.
8. t_{OPEN} is a measure of the break-before-make time.

*FIGURE 23. Switching time test circuit waveforms for table XVIII.

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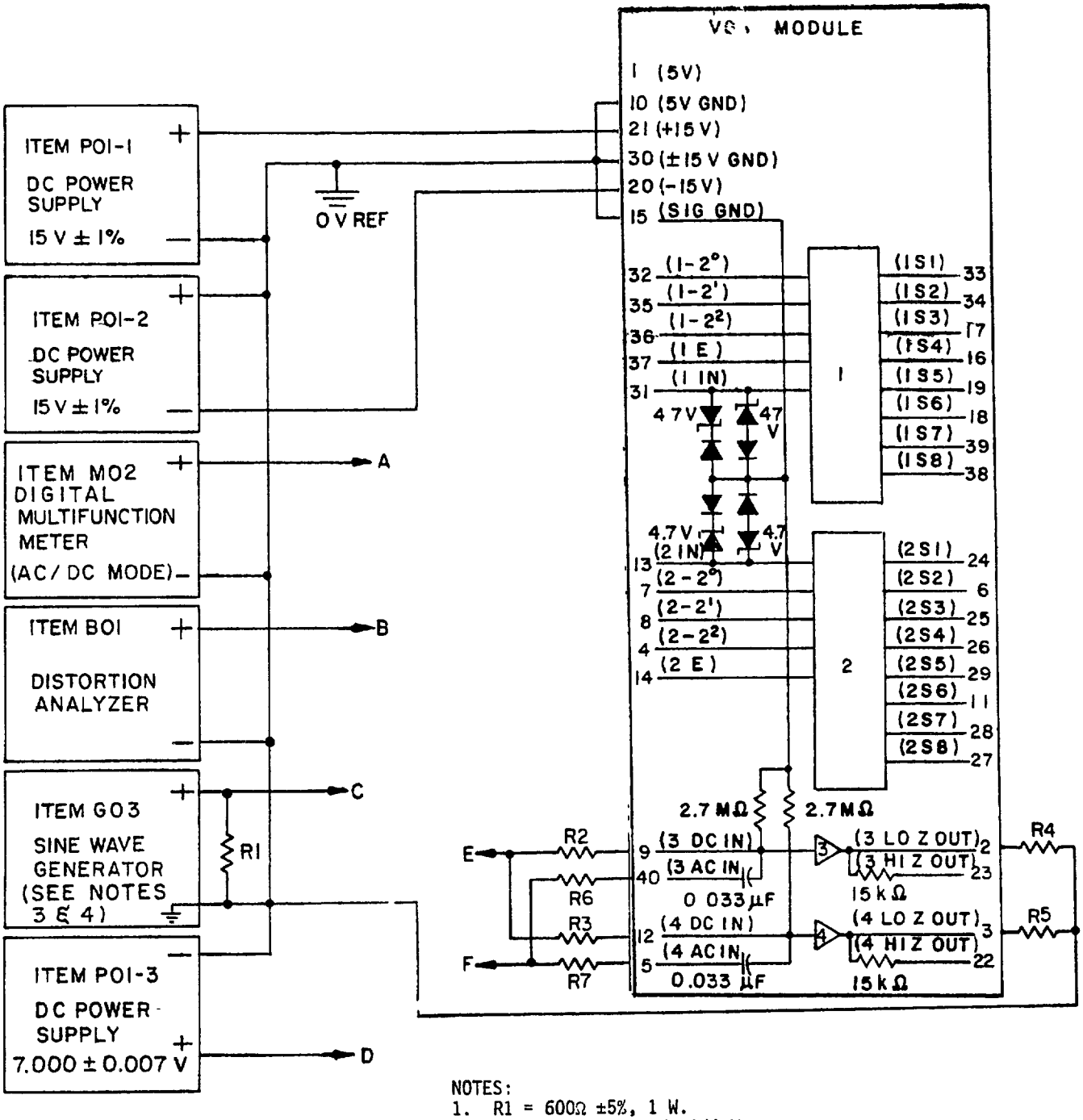
TABLE XIX. Large signal voltage response and distortion test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	3LO Z OUT Large signal voltage response	2	Open	F	Open	6.9	7.1	V rms	24
T2	4LO Z OUT Large signal voltage response	3	Open	"	"	6.9	7.1	V rms	"
T3	3LO Z OUT Distortion	Open	2	"	"	0	1	%	"
T4	4LO Z OUT Distortion	Open	3	"	"	0	1	%	"

TABLE XX. Low frequency response and DC gain test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	3LO Z OUT Low frequency response	2	Open	E	Open	6.93	7.07	V rms	24
U2	4LO Z OUT Low frequency response	3	"	E	Open	6.93	7.07	V rms	"
U3	3LO Z OUT DC gain	2	"	Open	E	6.94	7.06	V	"
U4	4LO Z OUT DC gain	3	"	Open	E	6.94	7.06	V	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T16:30Z
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NOTES:
1. R1 = 600Ω ±5%, 1 W.
2. R2-R7 = 10 kΩ ±1%, 1/4 W.
3. Item G03 setup for test conditions T1-T4.
   Amplitude = 7 V rms ±0.2%.
   Frequency = 20 kHz ±5%.
4. Item G03 setup for test conditions U1 and U2:
   Amplitude = 7 V rms ±0.2%.
   Frequency = 50 Hz ±5%.
5. Point E is open for test conditions T1-T4.
6. Point F is open for test conditions U1-U4.
7. Subtract input distortion from output distortion before comparing to table XIX limits.

* FIGURE 24. Large signal voltage response, distortion, low frequency response, and DC gain test circuit for tables XIX and XX.

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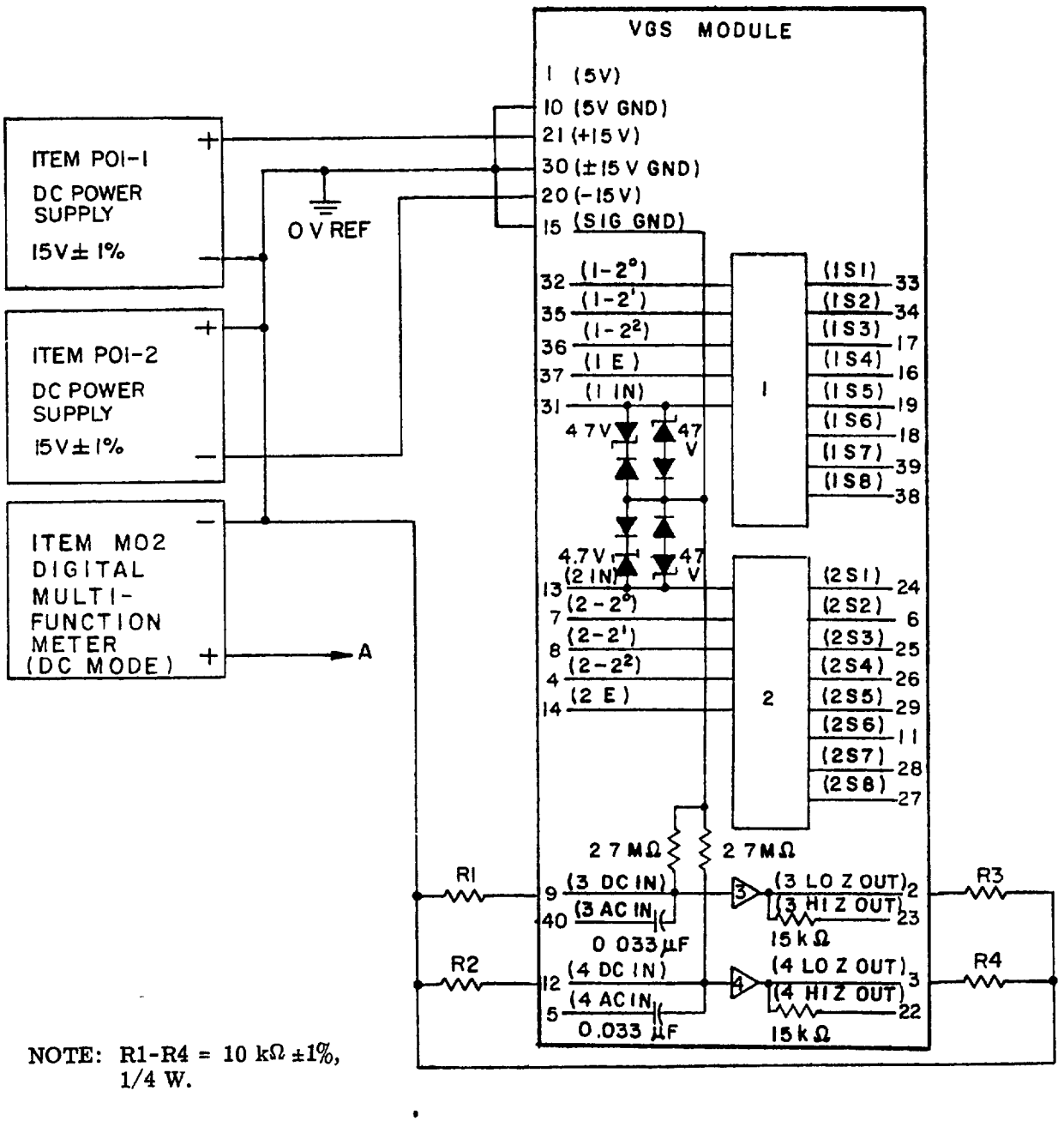
TABLE XXI. DC offset voltage test.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
V1	3LO Z OUT	DC offset voltage	2	-10	10	mV	25
V2	4LO Z OUT	DC offset voltage	3	-10	10	mV	25

TABLE XXII. Output resistance and input resistance test.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
W1	3HI Z OUT	Output resistance	23	2	Open	14.85	15.15	kΩ	26
W2	4HI Z OUT	Output resistance	22	3	Open	14.85	15.15	kΩ	"
W3	3DC IN	Input resistance	Open	Open	9	2.43	2.97	MΩ	"
W4	4DC IN	Input resistance	Open	Open	12	2.43	2.97	MΩ	"

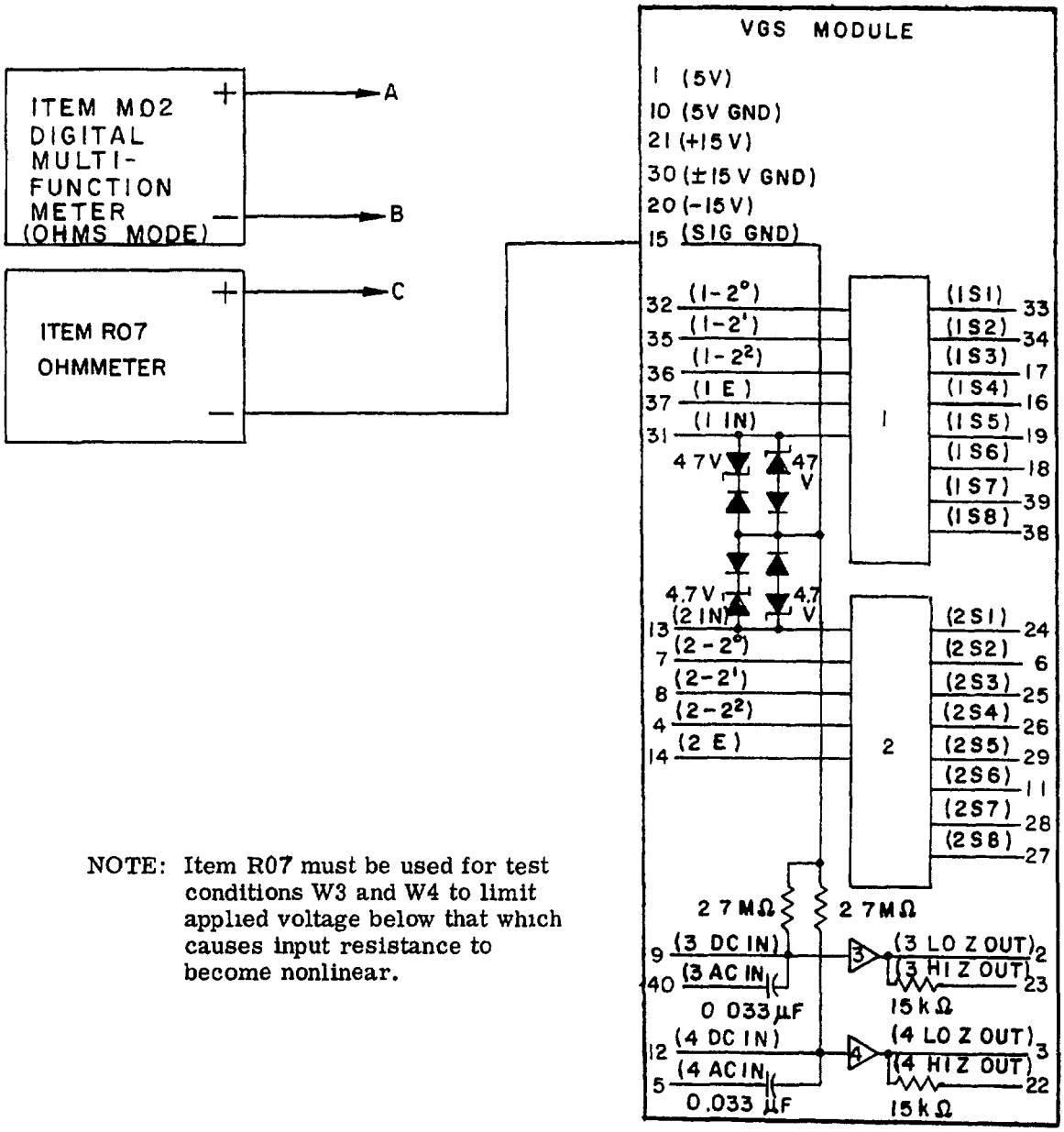
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NOTE: R1-R4 = 10 kΩ ±1%, 1/4 W.

*FIGURE 25. DC offset voltage test circuit for table XXI.

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NOTE: Item R07 must be used for test conditions W3 and W4 to limit applied voltage below that which causes input resistance to become nonlinear.

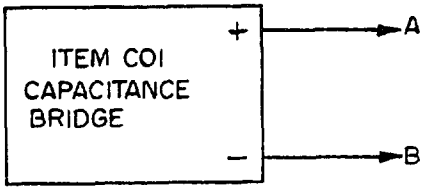
*FIGURE 26. Output resistance and input resistance test circuit for table XXII.

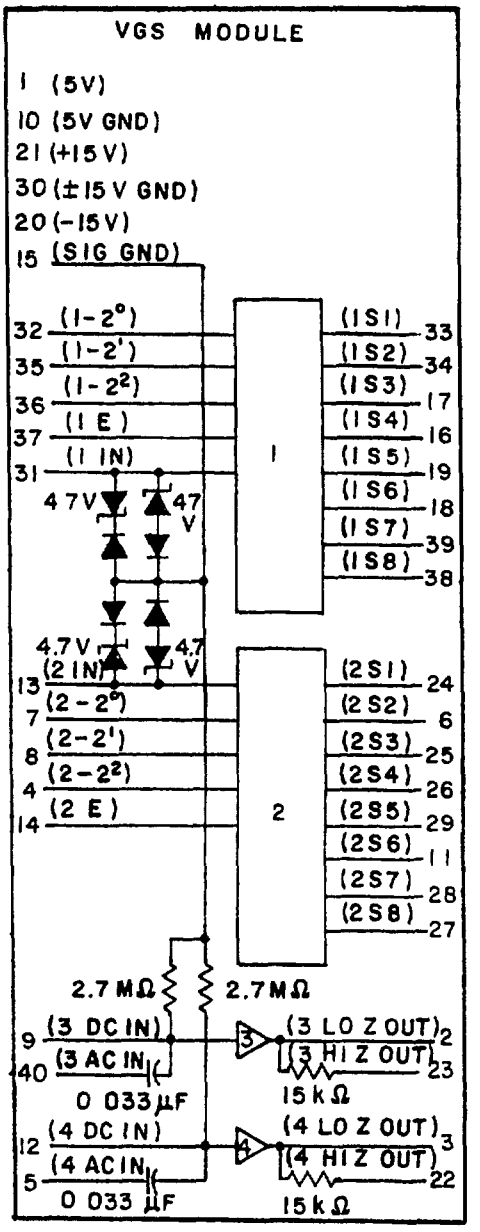
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TABLE XXIII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X1	Power supply filter capacitance	1	10	1.65	2.75	µF	27
X2	"	21	30	0.075	0.125	"	"
X3	"	30	20	0.075	0.125	"	"

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NOTES:
1. Null out the effect of the test leads prior to connecting the module.
2. Item C01 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

* FIGURE 27. Power supply filter capacitance test circuit for table XXIII.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/182A(EC)	Modules, Standard Electronic Demultiplexer, Analog, Key Code VGS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

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6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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